
Polynomial-Logarithmic Transseries

Algebra, Composition, Reversion, and the Lambert W Function

A self-contained computational and asymptotic treatment

Scope. We study expansions of the form

$$\sum_{n \geq n_0} p_n (\log x) x^{-n}, \quad p_n \in \mathbb{K}[X],$$

with special attention to the four operations that most often cause mistakes: multiplication, division, composition, and compositional inversion. The central running example is the large-argument expansion of the Lambert W function. The treatment distinguishes formal identities, closure properties, coefficient algorithms, and analytic asymptotic meaning.

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Abstract

Polynomial-logarithmic transseries are Laurent series in x^{-1} whose coefficients are polynomials in $\ell = \log x$. They form one of the smallest useful fragments of the logarithmic-exponential transseries field: small enough for exact coefficient arithmetic, yet rich enough to express the standard large-argument expansions of Lambert W , Wright's omega function, inverse logarithmic perturbations, and many solutions obtained from implicit equations.

This article develops the fragment as a complete filtered algebraic calculus. We define its rank-two monomial order and truncation topology; prove convolution, unit, differentiation, antidifferentiation, admissible-composition, and reversion theorems; derive explicit coefficient recurrences for products, quotients, near-identity compositions, and inverse series; and compare fixed-point, triangular, Newton, and Lagrange–Bürmann inversion algorithms. Particular care is given to operations that leave the polynomial-logarithmic ring. This leads naturally to rational-log, Laurent-log, Puiseux-log, and iterated-log enlargements.

For Lambert W , the article derives the complete expansion in Lambert polynomials, proves their differential-operator, integral-recursion, generating-function, and Stirling-number descriptions, treats complex branches and the real branch W_{-1} , and gives numerical and implementation guidance. Throughout, “formal”, “asymptotic”, and “convergent” are kept logically separate.

Source note. The supplied source bundle contains, among other material, G. A. Edgar's expositions of Hahn-series transseries and composition, papers on transseries and surreal/Hardy-field viewpoints, and several broad tutorials. The present article uses those materials as background but deliberately specializes to the rank-two polynomial-logarithmic class and develops coefficient-level algorithms not isolated in the broader surveys. Additional primary references include the NIST Digital Library of Mathematical Functions, Corless et al. on Lambert W , Cohen on Lambert polynomials, Gessel on formal Lagrange inversion, and van der Hoeven's monograph on transseries.

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1 Orientation: why this small transseries class matters

1.1 The characteristic shape

Let $x \rightarrow +\infty$, put

$$t = x^{-1}, \quad \ell = \log x,$$

and consider a formal expression

$$F(x) = \sum_{n \geq n_0} p_n(\ell) t^n = \sum_{n \geq n_0} p_n(\log x) x^{-n}, \quad p_n \in \mathbb{K}[X]. \quad (1.1)$$

Here $n_0 \in \mathbb{Z}$, so finitely many positive powers of x are permitted. At each fixed inverse-power level x^{-n} , only a finite polynomial in $\log x$ is allowed. We call (1.1) a *polynomial-logarithmic transseries*, abbreviated PL transseries.

The prototype is obtained by inverting

$$y + \log y = x.$$

The inverse is the Wright omega function $\omega(x) = W(e^x)$, and its expansion begins

$$\begin{aligned} \omega(x) = x - \ell + \frac{\ell}{x} + \frac{\ell^2/2 - \ell}{x^2} + \frac{\ell^3/3 - 3\ell^2/2 + \ell}{x^3} \\ + \frac{\ell^4/4 - 11\ell^3/6 + 3\ell^2 - \ell}{x^4} + \dots \end{aligned} \quad (1.2)$$

Every coefficient is a polynomial in ℓ . For $W(z)$, take $x = \text{Log } z$ and $\ell = \text{Log Log } z$, with branch choices recorded explicitly.

PL transseries occupy a useful middle ground.

- They are more expressive than ordinary Laurent series because logarithms are unavoidable under many asymptotic inversions.
- They are far simpler than unrestricted logarithmic-exponential transseries: a truncated PL series can be represented by a finite array of ordinary polynomials.
- Multiplication, differentiation, integration, many divisions, many compositions, and all near-identity reversions reduce to exact finite calculations at every requested order.
- Their failure modes are transparent. For example, $1/\ell$, $\log \ell$, and $x^{1/2}$ each indicate a specific minimal enlargement.

1.2 Four meanings that must not be conflated

The same displayed series can be read in several ways.

Definition 1.1 (Formal PL series). A formal PL series is an element of the algebra defined in [Section 2](#). Equality means equality of every coefficient polynomial; no value of x is substituted.

Definition 1.2 (Poincaré PL expansion). A function f has the expansion

$$f(x) \sim \sum_{n \geq n_0} p_n(\log x) x^{-n}$$

as $x \rightarrow +\infty$ if, after including all terms through inverse-power level $N - 1$, the remainder is $o(x^{-N+\varepsilon})$ for every fixed $\varepsilon > 0$; equivalently, it is smaller than every monomial at all retained levels and larger only at omitted levels. A sharper and usually more convenient formulation is given in [Section 9](#).

A formal series can also converge numerically for sufficiently large x , or be summable by an external analytic procedure. Neither follows merely from formal existence. Conversely, a convergent expansion is automatically formal when its coefficients are compared algebraically. Lambert W is unusually pleasant: its canonical inverse-logarithmic series is asymptotic and, in a sufficiently large domain, genuinely convergent.

1.3 What the article proves

The main algebraic conclusions may be summarized in advance.

Closure map

Let $\mathcal{P} = \mathbb{K}[\ell]((x^{-1}))$.

- (i) $\mathcal{P}$ is a differential algebra and is closed under indefinite integration up to an additive constant.
- (ii) It is closed under multiplication, with ordinary Cauchy convolution in the x^{-1} -index.
- (iii) It is not a field. A series is a unit precisely when its leading coefficient polynomial is a nonzero constant. General division belongs to $\mathbb{K}(\ell)((x^{-1}))$, and its large- ℓ expansion belongs to $\mathbb{K}((\ell^{-1}))((x^{-1}))$.
- (iv) If the inner argument is $G = cx^q(1 + U)$, where $q \in \mathbb{N}_{>0}$, $c \neq 0$, and $U \in x^{-1}\mathbb{K}[\ell][[x^{-1}]]$, then $F \circ G \in \mathcal{P}$ after choosing $\log c$. For rational q , one naturally obtains a Puiseux-log series.
- (v) The near-identity class $\mathcal{G} = \{x + A : A \in \mathbb{K}[\ell][[x^{-1}]]\}$ is a group under composition. Its inverse is computable by a contracting fixed point, triangular coefficient matching, Newton iteration, or a Lagrange–Bürmann formula.

2 The formal polynomial-logarithmic ring

2.1 Coefficient field and variables

Throughout most of the article, $\mathbb{K}$ is a field of characteristic zero. For real asymptotics one can take $\mathbb{K} = \mathbb{R}$; for branch-sensitive analytic work, $\mathbb{K} = \mathbb{C}$; and for exact symbolic computation, $\mathbb{K} = \mathbb{Q}$ or a finite algebraic extension is often ideal. The symbols x , $t = x^{-1}$, and $\ell = \log x$ are formal, but the derivation is chosen to reproduce their analytic relation:

$$\frac{dx}{dx} = 1, \quad \frac{dt}{dx} = -t^2, \quad \frac{d\ell}{dx} = t. \quad (2.1)$$

Definition 2.1 (PL Laurent ring). The polynomial-logarithmic Laurent ring is

$$\mathcal{P} = \mathbb{K}[\ell]((t)) = \left\{ \sum_{n \geq n_0} p_n(\ell) t^n : n_0 \in \mathbb{Z}, p_n \in \mathbb{K}[\ell] \right\}. \quad (2.2)$$

Its power-series subring is

$$\mathcal{P}_{\geq 0} = \mathbb{K}[\ell][[t]],$$

and its strictly small ideal is

$$\mathfrak{m} = t\mathbb{K}[\ell][[t]].$$

The notation is an *iterated* Laurent-series notation: first form the polynomial coefficient ring $\mathbb{K}[\ell]$, then form Laurent series in t . This choice records that inverse powers of x are the primary asymptotic levels and powers of $\log x$ refine each level.

2.2 Rank-two monomial order

The monomials are

$$t^n \ell^j = x^{-n} (\log x)^j, \quad n \in \mathbb{Z}, \quad j \in \mathbb{N}.$$

For positive real $x \rightarrow \infty$, define dominance by

$$t^n \ell^j \succ t^m \ell^k \iff n < m \quad \text{or} \quad (n = m \text{ and } j > k). \quad (2.3)$$

Thus every power of x dominates every power of $\log x$:

$$x \succ \ell^{1000} \succ 1 \succ \ell^{-1000} \succ x^{-1}.$$

Within $\mathcal{P}$, negative powers of ℓ are absent, but they will appear in the field extension $\mathcal{H}$ below.

It is often convenient to encode (2.3) by the rank-two valuation

$$v(t^n \ell^j) = (n, -j) \in \mathbb{Z} \times \mathbb{Z} \quad (2.4)$$

with lexicographic order. A nonzero $F \in \mathcal{P}$ has a least valuation because its t -exponents are bounded below and the polynomial at the first nonzero level has finite degree.

Definition 2.2 (Leading data). For nonzero $F = \sum_{n \geq n_0} p_n(\ell) t^n$, let

$$\text{ord}_{x^{-1}} F = \min\{n : p_n \neq 0\}, \quad d(F) = \deg_{\ell} p_{\text{ord}_{x^{-1}} F}.$$

If $a \ell^{d(F)}$ is the leading term of $p_{\text{ord}_{x^{-1}} F}$, then

$$\text{lm}(F) = t^{\text{ord}_{x^{-1}} F} \ell^{d(F)}, \quad \text{lc}(F) = a.$$

The pair $(\text{ord}_{x^{-1}} F, -d(F))$ is the valuation of F . Notice that $\text{ord}_{x^{-1}}$ alone is only a coarse valuation: two terms at the same inverse-power level are distinguished by their log degree.

2.3 Why all coefficient sums are finite

The support of F is

$$\text{supp } F = \{(n, j) : [t^n \ell^j] F \neq 0\}.$$

For every fixed n , its fiber in j is finite, and the set of occurring n 's is bounded below. Consequently every nonempty subset of $\text{supp } F$ has a dominant monomial. This is the elementary rank-two instance of the well-based support condition used for Hahn series and general transseries [5, 8].

For multiplication, a fixed pair (r, k) can arise only from $n + m = r$, and there are finitely many such pairs once lower bounds on n and m are fixed. At each such pair, polynomial multiplication supplies only finitely many log-degree contributions. This local finiteness is the basic reason that all operations below can be performed to finite order.

2.4 Truncation and completeness

For $N \in \mathbb{Z}$, define the inverse-power truncation

$$\mathbf{T}_{<N}(F) = \sum_{n < N} p_n(\ell) t^n, \quad F \equiv G \pmod{t^N} \iff F - G \in t^N \mathbb{K}[\ell][[t]]. \quad (2.5)$$

A computation “through order x^{-N+1} ” means modulo t^N . This convention retains *all* powers of ℓ at each retained t -level. It avoids the common ambiguity of a total-degree cutoff in two variables, which generally does not respect asymptotic dominance.

The metric

$$d(F, G) = 2^{-\text{ord}_{x^{-1}}(F-G)}$$

(on differences with nonnegative t -order, with the usual convention for zero) defines the t -adic topology. A sequence converges exactly when every coefficient polynomial eventually stabilizes. The ring $\mathbb{K}[\ell][[t]]$ is complete in this topology. This elementary completeness supports geometric series, fixed-point iteration, and Newton reversion.

Remark 2.3 (Log-degree truncation is a separate approximation). A computer implementation may also cap polynomial degree in ℓ , but that is not part of the intrinsic t -adic topology. Unless a theorem supplies a degree bound, silently dropping high log powers can corrupt coefficients at an already retained inverse-power level.

2.5 Degree profiles

For $F = \sum p_n(\ell)t^n$, define

$$d_F(n) = \begin{cases} \deg_\ell p_n, & p_n \neq 0, \\ -\infty, & p_n = 0. \end{cases}$$

The sequence $d_F(n)$ is the *log-degree profile*. In many inverse problems it is at most affine in n . For the Lambert expansion, $d_F(n) = n$ for $n \geq 1$. Tracking the profile provides a sharper memory and complexity estimate than treating every coefficient polynomial as having one global maximum degree.

3 Multiplication and division

3.1 Cauchy convolution over $\mathbb{K}[\ell]$

Write

$$F = \sum_{n \geq a} p_n(\ell)t^n, \quad G = \sum_{n \geq b} q_n(\ell)t^n.$$

Theorem 3.1 (Product formula). *The product $FG \in \mathcal{P}$ is*

$$FG = \sum_{r \geq a+b} r_r(\ell)t^r, \quad r_r(\ell) = \sum_{n+m=r} p_n(\ell)q_m(\ell). \quad (3.1)$$

Every coefficient sum in (3.1) is finite. Moreover,

$$\text{ord}_{x^{-1}}(FG) = \text{ord}_{x^{-1}} F + \text{ord}_{x^{-1}} G, \quad d(FG) = d(F) + d(G), \quad \text{lc}(FG) = \text{lc}(F) \text{lc}(G).$$

Proof. For fixed r , the bounds $n \geq a$ and $m \geq b$, together with $n + m = r$, imply $a \leq n \leq r - b$; hence only finitely many n occur. Each summand is a polynomial in ℓ , proving closure. At the first possible level $r = a + b$, only $(n, m) = (a, b)$ contributes, so the leading coefficient polynomial is $p_a q_b$. The assertions about order, degree, and leading scalar follow from the corresponding facts for ordinary polynomial products. $\square$

The degree profile satisfies the max-plus bound

$$d_{FG}(r) \leq \max_{n+m=r} (d_F(n) + d_G(m)). \quad (3.2)$$

Equality can fail through cancellation. If $d_F(n) \leq \alpha n + \beta$ and $d_G(n) \leq \alpha n + \gamma$ over the relevant ranges, then $d_{FG}(r) \leq \alpha r + \beta + \gamma$. Thus a common affine slope is stable under multiplication.

A complete finite convolution

Let

$$F = 1 + \ell t + (\ell^2 + 1)t^2, \quad G = 2 - \ell t + 3\ell t^2.$$

Then

$$\begin{aligned} FG &= 2 + \ell t + (\ell^2 + 3\ell + 2)t^2 \\ &\quad + (-\ell^3 + 3\ell^2 - \ell)t^3 + (3\ell^3 + 3\ell)t^4. \end{aligned}$$

The calculation is ordinary polynomial multiplication nested inside ordinary Cauchy convolution.

3.2 Truncated multiplication

Suppose only $FG \bmod t^N$ is required. Terms of F with exponent $n \geq N - b$ and terms of G with exponent $m \geq N - a$ cannot contribute. Hence

$$FG \bmod t^N = \sum_{r=a+b}^{N-1} \left(\sum_{n=\max(a, r-(N_G-1))}^{\min(N_F-1, r-b)} p_n q_{r-n} \right) t^r, \quad (3.3)$$

where N_F, N_G denote the available cutoffs. A robust implementation carries lower exponents explicitly rather than pretending every series starts at zero.

3.3 Units in the polynomial-logarithmic ring

The distinction between a nonzero element and a unit is essential.

Theorem 3.2 (Unit criterion). *Let*

$$G = t^m (g_0(\ell) + g_1(\ell)t + g_2(\ell)t^2 + \cdots), \quad g_0 \neq 0.$$

Then G is invertible in $\mathcal{P}$ if and only if $g_0 \in \mathbb{K}^\times$, that is, the leading coefficient polynomial is a nonzero constant.

Proof. If $g_0 = c \in \mathbb{K}^\times$, factor

$$G = ct^m(1 + U), \quad U \in t\mathbb{K}[\ell][[t]],$$

and use the t -adically convergent geometric series

$$G^{-1} = c^{-1}t^{-m} \sum_{r \geq 0} (-U)^r. \quad (3.4)$$

At any fixed t -order only finitely many powers of U contribute.

Conversely, suppose $GH = 1$ with $H \in \mathcal{P}$. Comparing leading t -levels shows that the leading coefficient polynomials of G and H multiply to 1 in $\mathbb{K}[\ell]$. The only units of a polynomial ring over a field are nonzero constants, so g_0 must be constant. $\square$

A frequent category error

The ring $\mathbb{K}[\ell]((t))$ is not a field. For example,

$$\ell^{-1} \notin \mathcal{P}, \quad (\ell + t)^{-1} \notin \mathcal{P}.$$

Calling every nonzero PL series “invertible” silently changes the coefficient ring. The appropriate extensions are described in [Section 3.6](#).

3.4 Reciprocal recurrence

Let

$$G = \sum_{n \geq 0} g_n(\ell) t^n, \quad g_0 = c \in \mathbb{K}^\times, \quad G^{-1} = \sum_{n \geq 0} b_n(\ell) t^n.$$

Comparing coefficients in $GG^{-1} = 1$ gives

$$b_0 = c^{-1}, \quad b_n = -c^{-1} \sum_{k=1}^n g_k b_{n-k} \quad (n \geq 1). \quad (3.5)$$

This recurrence is usually preferable to expanding the geometric series literally: it computes each new coefficient once and exposes exact polynomial cancellation.

Example 3.3. For $G = 1 + \ell t + t^2$, (3.5) yields

$$\begin{aligned} G^{-1} &= 1 - \ell t + (\ell^2 - 1)t^2 + (-\ell^3 + 2\ell)t^3 \\ &\quad + (\ell^4 - 3\ell^2 + 1)t^4 + (-\ell^5 + 4\ell^3 - 3\ell)t^5 + \dots \end{aligned}$$

The coefficient polynomials obey the second-order recurrence $b_n = -\ell b_{n-1} - b_{n-2}$.

3.5 Long division and quotient recurrence

Suppose

$$F = \sum_{n \geq 0} f_n t^n, \quad G = \sum_{n \geq 0} g_n t^n, \quad g_0 = c \in \mathbb{K}^\times, \quad H = F/G = \sum_{n \geq 0} h_n t^n.$$

Then

$$h_0 = c^{-1} f_0, \quad h_n = c^{-1} \left(f_n - \sum_{k=1}^n g_k h_{n-k} \right). \quad (3.6)$$

The same formula works for Laurent series after first shifting the leading powers of t . It is triangular: h_n depends only on earlier quotient coefficients.

If $g_0(\ell)$ is nonconstant, the same recurrence is valid in $\mathbb{K}(\ell)$, and it is also an exact divisibility test in $\mathbb{K}[\ell]$: at every step the numerator must be divisible by g_0 . A quotient can therefore lie in $\mathcal{P}$ even when G is not a unit, but this is exceptional and must be verified coefficient by coefficient.

3.6 The minimal field extensions

Three nested coefficient choices are useful:

$$\mathcal{P} = \mathbb{K}[\ell]((t)), \quad (3.7)$$

$$\mathcal{R} = \mathbb{K}(\ell)((t)), \quad (3.8)$$

$$\mathcal{H} = \mathbb{K}((\ell^{-1}))((t)). \quad (3.9)$$

Here $\mathcal{R}$ is a field of Laurent series in t with rational functions of ℓ as coefficients. The field $\mathcal{H}$ expands each rational-log coefficient at $\ell = \infty$, allowing finitely many positive and arbitrarily many negative powers of ℓ . For example,

$$\begin{aligned} \frac{1}{\ell + t} &= \frac{1}{\ell} \frac{1}{1 + t/\ell} = \ell^{-1} - t\ell^{-2} + t^2\ell^{-3} - \dots \in \mathcal{H}, \\ \frac{1}{\ell^2 + \ell + 1} &= \ell^{-2} \left(1 + \ell^{-1} + \ell^{-2} \right)^{-1} \in \mathbb{K}((\ell^{-1})). \end{aligned}$$

Proposition 3.4 (Field property). *Both $\mathcal{R}$ and $\mathcal{H}$ are fields. Every nonzero element of $\mathcal{P}$ becomes invertible in $\mathcal{R}$, and its large- ℓ reciprocal expansion belongs to $\mathcal{H}$.*

Proof. A Laurent-series ring $E((t))$ over a field E is a field: factor a nonzero series as $t^m a(1 + U)$, where $a \in E^\times$ and $U \in tE[[t]]$, then use (3.4). Take $E = \mathbb{K}(\ell)$ or $E = \mathbb{K}((\ell^{-1}))$. A nonzero polynomial in ℓ is invertible in $\mathbb{K}(\ell)$; factoring its leading power of ℓ and geometrically expanding the lower-degree part gives an element of $\mathbb{K}((\ell^{-1}))$. $\square$

The order of iteration matters. The fields $\mathbb{K}((\ell^{-1}))((t))$ and $\mathbb{K}((t))((\ell^{-1}))$ impose different support conditions, just as iterated Laurent-series fields in ordinary formal-power-series theory are generally different [7]. Our choice makes $t = x^{-1}$ the primary scale, matching the intended asymptotic hierarchy.

3.7 Local analytic functional calculus

Let $U \in \mathfrak{m}$, and let $\Phi(z) = \sum_{r \geq 0} c_r (z - a)^r$ be a formal or convergent power series about $a \in \mathbb{K}$. Then

$$\Phi(a + U) = \sum_{r \geq 0} c_r U^r \in \mathbb{K}[\ell][[t]] \quad (3.10)$$

is well-defined t -adically. In particular,

$$(1 + U)^\alpha = \sum_{r \geq 0} \binom{\alpha}{r} U^r, \quad (3.11)$$

$$\log(1 + U) = \sum_{r \geq 1} \frac{(-1)^{r+1}}{r} U^r, \quad (3.12)$$

$$e^U = \sum_{r \geq 0} \frac{U^r}{r!}. \quad (3.13)$$

These formulas are formal; analytic convergence for a numerical x is a separate issue.

The restrictions are important. If $P(\ell)$ is a nonlinear polynomial, then $e^{P(\ell)}$ is generally not a PL series: for example, $e^{\ell^2} = x^\ell$ introduces a new exponential monomial. By contrast, $e^{a\ell+b} = e^{b_x} x^a$ remains in a generalized-power extension, and remains in $\mathcal{P}$ when $a \in \mathbb{Z}$.

4 Differential and integral calculus

4.1 The basic lowering operator on coefficient polynomials

For a monomial level $t^n p(\ell) = x^{-n} p(\log x)$, the chain rule gives

$$\frac{d}{dx} (t^n p(\ell)) = t^{n+1} (\partial_\ell - n) p(\ell). \quad (4.1)$$

Thus differentiation raises the t -order by one and applies a first-order operator to the log polynomial.

Definition 4.1 (Shifted logarithmic differential operator). For $n \in \mathbb{Z}$ and $k \in \mathbb{N}$, set

$$\Delta_{n,k} = \prod_{j=0}^{k-1} (\partial_\ell - n - j), \quad \Delta_{n,0} = 1. \quad (4.2)$$

The factors commute because each is a polynomial in ∂_ℓ .

Proposition 4.2 (Repeated derivative). *For every $k \geq 0$,*

$$\frac{d^k}{dx^k}(t^n p(\ell)) = t^{n+k} \Delta_{n,k} p(\ell). \quad (4.3)$$

Consequently $\mathcal{P}$ is closed under differentiation.

Proof. The case $k = 0$ is immediate and the case $k = 1$ is (4.1). If (4.3) holds for k , apply (4.1) at level $n + k$:

$$D(t^{n+k} \Delta_{n,k} p) = t^{n+k+1} (\partial_\ell - n - k) \Delta_{n,k} p = t^{n+k+1} \Delta_{n,k+1} p.$$

Termwise differentiation is locally finite at every t -level. $\square$

A useful expanded form follows from the signed Stirling numbers of the first kind. Since

$$\prod_{j=0}^{k-1} (z - n - j) = (z - n)^{\underline{k}},$$

$\Delta_{n,k}$ is a degree- k polynomial in ∂_ℓ . Because differentiation of a polynomial is nilpotent, only finitely many terms act nontrivially on any fixed coefficient.

4.2 Valuation behavior

For nonzero $F \in \mathcal{P}$, one always has

$$\text{ord}_{x^{-1}} F' \geq \text{ord}_{x^{-1}} F + 1.$$

Equality can fail. At level $n = 0$, a constant coefficient differentiates to zero; more generally, cancellations may annihilate the leading coefficient. For a leading term $at^n \ell^d$:

- if $n \neq 0$, then F' has leading term $-nat^{n+1} \ell^d$;
- if $n = 0$ and $d > 0$, then it has leading term $adt \ell^{d-1}$;
- if $n = d = 0$, the leading constant disappears and the next term decides.

This distinction is central in the contraction proof for compositional inverses: if $A \in \mathbb{K}[\ell][[t]]$, then $A' \in t\mathbb{K}[\ell][[t]]$.

4.3 Antidifferentiation stays in the class

Unlike many restricted asymptotic scales, $\mathcal{P}$ is also closed under indefinite integration.

Theorem 4.3 (Termwise antiderivative). *For every $F \in \mathcal{P}$, there exists $G \in \mathcal{P}$ with $G' = F$. It is unique up to an additive constant in $\mathbb{K}$.*

Proof. It suffices to integrate one term $t^n p(\ell) = x^{-n} p(\ell)$. If $n \neq 1$, seek an antiderivative of the form $x^{1-n} q(\ell) = t^{n-1} q(\ell)$. Differentiation gives

$$\frac{d}{dx}(x^{1-n} q(\ell)) = x^{-n} (q'(\ell) + (1-n)q(\ell)).$$

The operator $\partial_\ell + (1-n)$ is invertible on $\mathbb{K}[\ell]$ when $1-n \neq 0$: solve recursively from the highest polynomial degree downward. If $n = 1$, then

$$\int x^{-1} p(\log x) dx = \int p(\ell) d\ell,$$

which is again a polynomial in ℓ because $\text{char } \mathbb{K} = 0$. Termwise integration is locally finite, and the only kernel of the derivation in $\mathcal{P}$ is $\mathbb{K}$. $\square$

For $n \neq 1$, an explicit finite formula is

$$(\partial_\ell + 1 - n)^{-1} p = \frac{1}{1 - n} \sum_{j=0}^{\deg p} \left(-\frac{\partial_\ell}{1 - n} \right)^j p. \quad (4.4)$$

The sum terminates because sufficiently many derivatives of p vanish.

4.4 Product, quotient, and chain rules

Because all operations are defined coefficientwise in a complete filtered algebra, the usual formal rules hold whenever the relevant expression exists:

$$(FG)' = F'G + FG', \quad (4.5)$$

$$(F/G)' = (F'G - FG')/G^2, \quad (4.6)$$

$$(F \circ G)' = (F' \circ G)G'. \quad (4.7)$$

For the chain rule, admissibility of the composition must first be established; this is the subject of [Section 5](#). The algebraic identities can then be proved on finite truncations and passed to the t -adic limit.

5 Composition

5.1 Why substitution needs an admissibility theorem

For polynomials, substitution is automatic. For an infinite series it is not. If one writes

$$\left(\sum_n a_n M_n \right) \circ G = \sum_n a_n (M_n \circ G),$$

one must prove that only finitely many summands contribute to each output monomial. Otherwise a single coefficient would be an undefined infinite sum in $\mathbb{K}$. General transseries theory formulates this requirement through summable families and well-based supports [5, 6, 10]. In the PL setting it reduces to concrete order estimates.

Three examples show why a blanket rule “replace x by G ” is unsafe:

$$\begin{aligned} (x^{-1}) \circ \ell &= \ell^{-1} \notin \mathcal{P}, \\ (\log x) \circ (x\ell) &= \ell + \log \ell \notin \mathcal{P}, \\ (x^{-1}) \circ x^{1/2} &= x^{-1/2}, \end{aligned}$$

where the last expression lies in a Puiseux extension rather than in $\mathcal{P}$. The substitutions are meaningful in larger transseries fields, but not closed in the original ring.

5.2 Composition with a monomial-leading argument

The largest natural class of inner arguments that preserves integer-power PL series has a positive integral power of x as its leading monomial.

Theorem 5.1 (Monomial-leading composition). *Let*

$$F(y) = \sum_{n \geq a} p_n (\log y) y^{-n} \in \mathcal{P} \quad (5.1)$$

and let

$$G(x) = cx^q(1 + U(x)), \quad c \in \mathbb{K}^\times, \quad q \in \mathbb{N}_{>0}, \quad U \in \mathfrak{m}. \quad (5.2)$$

Choose a value of $\log c$ if $\mathbb{K} \subseteq \mathbb{C}$. Then

$$(F \circ G)(x) = \sum_{n \geq a} p_n(q\ell + \log c + V) c^{-n} t^{qn} (1 + U)^{-n}, \quad V = \log(1 + U), \quad (5.3)$$

is a well-defined element of $\mathcal{P}$.

Proof. By (3.12), $V \in \mathfrak{m}$. Since p_n is a polynomial, $p_n(q\ell + \log c + V)$ is a finite sum of powers of V , hence belongs to $\mathbb{K}[\ell][[t]]$. The binomial expansion

$$(1 + U)^{-n} = \sum_{r \geq 0} \binom{-n}{r} U^r$$

is t -adically well-defined for every integer n . The n -th outer summand begins at t^{qn} , up to the fixed finite set of negative n 's. For a requested output exponent N , only $n \leq N/q$ can contribute. For each such n , only finitely many powers of U and V can contribute modulo t^{N+1} . Thus every coefficient is a finite sum of polynomials in ℓ , proving both summability and closure. $\square$

Remark 5.2 (Rational and general powers). If $q = r/s \in \mathbb{Q}_{>0}$, the same proof works after adjoining $x^{1/s}$, giving a Puiseux-log series. For arbitrary real or complex q , one enters a generalized Hahn-series scale with monomials $x^{-\gamma}$. The obstruction is not composition itself; it is the decision to restrict exponents to integers.

Corollary 5.3 (Scaling and power substitution). *For $c \neq 0$ and $q \in \mathbb{N}_{>0}$,*

$$F(cx^q) = \sum_{n \geq a} c^{-n} x^{-qn} p_n(q\ell + \log c). \quad (5.4)$$

In particular, scaling acts on coefficient polynomials by an affine change of logarithmic variable.

5.3 Near-identity arguments

The most important special case is

$$G(x) = x + H(x) = x(1 + tH(x)), \quad H \in \mathcal{P}_{\geq 0}. \quad (5.5)$$

Since $tH \in \mathfrak{m}$, [Theorem 5.1](#) applies with $q = 1$, $c = 1$, and $U = tH$. Analytically, $H = o(x)$, even when H is a polynomial in $\log x$.

Theorem 5.4 (Formal Taylor formula). *For $F \in \mathcal{P}$ and $H \in \mathcal{P}_{\geq 0}$,*

$$F \circ (x + H) = \sum_{k \geq 0} \frac{H^k}{k!} D^k F, \quad D = \frac{d}{dx}. \quad (5.6)$$

The family on the right is t -adically summable. Modulo t^N , only finitely many k are needed.

Proof. Introduce an auxiliary central variable s and form $\Phi(s) = F(x + sH)$ using [Theorem 5.1](#). Differentiation with respect to s , holding x and $H(x)$ fixed, gives

$$\frac{\partial^k \Phi}{\partial s^k}(0) = H^k D^k F.$$

The formal Taylor expansion in s therefore gives $\Phi(s) = \sum_{k \geq 0} s^k H^k D^k F / k!$. Set $s = 1$. If $a = \text{ord}_{x^{-1}} F$, then $D^k F$ has t -order at least $a + k$, while H^k has nonnegative t -order. Hence at output order N , only $k \leq N - a$ can contribute. This proves summability and justifies setting $s = 1$. $\square$

The theorem is an exact identity of formal PL series, not merely an asymptotic Taylor approximation. Its validity rests on the filtration: successive Taylor terms become successively smaller in powers of t .

5.4 An explicit coefficient formula

Let

$$F = \sum_{n \geq a} p_n(\ell) t^n, \quad H^k = \sum_{m \geq 0} h_{k,m}(\ell) t^m, \quad h_{0,0} = 1. \quad (5.7)$$

Combining Theorems 4.2 and 5.4 yields

$$[t^r] F \circ (x + H) = \sum_{\substack{n \geq a, k \geq 0, m \geq 0 \\ n+k+m=r}} \frac{h_{k,m}(\ell)}{k!} \Delta_{n,k} p_n(\ell). \quad (5.8)$$

The sum is finite. Formula (5.8) separates the three pieces of work:

1. powers H^k , computed by truncated multiplication;
2. shifted differential operators $\Delta_{n,k}$ on coefficient polynomials;
3. a finite index convolution $n + k + m = r$.

It is particularly convenient when the same inner correction H is used to compose many outer series.

Composition by $x + \ell$

Let

$$F(x) = \ell + \ell^2 t, \quad G(x) = x + \ell.$$

Directly,

$$\log G = \ell + \log(1 + \ell t), \quad G^{-1} = t(1 + \ell t)^{-1}.$$

Either these formulas or (5.8) gives

$$\begin{aligned} F \circ G &= \ell + (\ell^2 + \ell)t + \left(-\ell^3 + \frac{3}{2}\ell^2\right)t^2 \\ &\quad + \left(\ell^4 - \frac{8}{3}\ell^3 + \ell^2\right)t^3 \\ &\quad + \left(-\ell^5 + \frac{41}{12}\ell^4 - 2\ell^3\right)t^4 + O(t^5). \end{aligned}$$

The increasing log degree is not accidental: every expansion of $\log(1 + \ell t)$ supplies another factor of ℓ .

5.5 Composition by an affine perturbation

For $G = cx + d + O(x^{-1})$, factor

$$G = cx \left(1 + \frac{d}{cx} + O(x^{-2})\right).$$

Then

$$\begin{aligned} \log G &= \ell + \log c + \frac{d}{c}t - \frac{d^2}{2c^2}t^2 + \cdots, \\ G^{-n} &= c^{-n}t^n \left(1 + \frac{d}{c}t + \cdots\right)^{-n}. \end{aligned}$$

The constant shift d is therefore handled by the same binomial-logarithmic calculus as any near-identity correction. There is no need to introduce a separate theory of “series at infinity” beyond the filtered ring already in use.

5.6 Associativity and the chain rule

Composition is associative only when all displayed compositions are admissible. Within the monomial-leading domain of [Theorem 5.1](#), the identity

$$(F \circ G) \circ H = F \circ (G \circ H) \quad (5.9)$$

holds whenever $G \circ H$ again has an admissible positive large leading monomial. One proof checks the identity first for the generators x , x^{-1} , and $\log x$, then extends through addition, multiplication, and t -adic limits. Another truncates all series, where substitution is a finite algebraic operation, and lets the cutoff increase.

Under the same hypotheses,

$$D(F \circ G) = (DF \circ G)DG. \quad (5.10)$$

For near-identity $G = x + H$, it can also be verified by differentiating [\(5.6\)](#) and rearranging the locally finite sums.

5.7 Exactly where closure fails

The following table is a practical diagnostic.

Inner leading behavior	Typical new object	Minimal customary enlargement
$cx^q, q \in \mathbb{N}_{>0}$	none	$\mathcal{P}$
$cx^{r/s}$	fractional powers of x	Puiseux-log series
$c\ell^q$	negative powers of ℓ from outer reciprocals	Laurent-log field $\mathcal{H}$
$x\ell^a$ inside an outer logarithm	$\log \ell$	one additional iterated-log level
e^x	exponential monomials	logarithmic-exponential transseries
inner series near a branch zero	fractional or ramified local powers	local Puiseux/branch expansion

Warning 5.5. An expression can be meaningful while lying outside $\mathcal{P}$. Closure is a statement about the chosen representation class, not about existence of the composed analytic function.

6 Compositional inversion and series reversal

6.1 The near-identity composition group

Definition 6.1 (Near-identity PL maps). Let

$$\mathcal{G} = \{x + A(x) : A \in \mathcal{P}_{\geq 0}\}. \quad (6.1)$$

Every member satisfies $x + A \sim x$ and has formal derivative $1 + A' = 1 + O(t)$.

Theorem 6.2 (Group theorem). *The set $\mathcal{G}$ is a group under composition. If $F = x + A \in \mathcal{G}$, there is a unique $G = x + B \in \mathcal{G}$ such that*

$$F \circ G = G \circ F = x. \quad (6.2)$$

Proof. Closure follows from

$$(x + A) \circ (x + B) = x + B + A \circ (x + B),$$

and [Theorem 5.4](#) shows that the correction belongs to $\mathcal{P}_{\geq 0}$. For existence, solve the fixed-point equation

$$B = \Phi(B) := -A \circ (x + B). \quad (6.3)$$

If $B - C \in t^N \mathcal{P}_{\geq 0}$, the formal mean-value/Taylor identity gives

$$A \circ (x + B) - A \circ (x + C) = (B - C) \int_0^1 A' \circ (x + C + s(B - C)) \, ds.$$

Since $A' \in t \mathcal{P}_{\geq 0}$, the right side lies in $t^{N+1} \mathcal{P}_{\geq 0}$. Thus Φ improves the agreement order by at least one. Starting with $B_0 = 0$, the iterates $B_{j+1} = \Phi(B_j)$ form a Cauchy sequence in the complete t -adic ring and converge to a unique fixed point B . The equation $B + A \circ (x + B) = 0$ gives $F \circ G = x$.

A right inverse in this class is automatically two-sided. Indeed, construct a right inverse H of G . Then

$$F = F \circ x = F \circ (G \circ H) = (F \circ G) \circ H = x \circ H = H,$$

so $G \circ F = G \circ H = x$. Uniqueness follows from the contraction: two fixed points agree modulo t^N for all N . $\square$

This is the PL specialization of the fixed-point mechanism used for well-based transseries compositional inverses [6]. Here the contraction is visible in the elementary fact $A' = O(t)$.

6.2 Picard iteration

The proof supplies the first inversion algorithm:

$$B_0 = 0, \quad B_{j+1} = -A \circ (x + B_j). \quad (6.4)$$

After j iterations, at least the first j new inverse-power levels have stabilized (often more, if A' begins beyond order t). This method is conceptually simple and excellent for proofs, but it repeatedly recomputes lower coefficients and is not the fastest high-order algorithm.

6.3 Triangular coefficient correction

Suppose an approximation $G_N = x + B_N$ satisfies

$$F \circ G_N - x = r_N(\ell)t^N + O(t^{N+1}). \quad (6.5)$$

Because $F' \circ G_N = 1 + O(t)$, adding $-r_N t^N$ to G_N removes the residual at level N :

$$G_{N+1} = G_N - r_N(\ell)t^N \implies F \circ G_{N+1} = x + O(t^{N+1}). \quad (6.6)$$

Thus inverse coefficients can be found one polynomial at a time. No linear system larger than a scalar identity is needed because the leading derivative is one. For a more general leading slope $F(x) = ax + o(x)$, the correction is divided by a .

6.4 Newton reversion

The formal Newton step is

$$G_{\text{new}} = G - \frac{F \circ G - x}{F' \circ G}. \quad (6.7)$$

The denominator is a unit because $F' = 1 + O(t)$. If $F \circ G - x \in t^N \mathcal{P}_{\geq 0}$, Taylor expansion around G shows that the new residual is in $t^{2N+1} \mathcal{P}_{\geq 0}$ (the exact exponent can improve with additional valuation information). Hence a precision-doubling implementation uses truncation cutoffs 1, 2, 4, 8, $\dots$, recomputing products, composition, and a reciprocal only at the newly required precision.

Residual estimate. Let $R = F \circ G - x$ and $C = -R/(F' \circ G)$. Since $C = O(t^N)$,

$$F \circ (G + C) - x = R + (F' \circ G)C + \frac{1}{2}(F'' \circ G)C^2 + \dots$$

The first two terms cancel. Now $F'' = A'' = O(t^2)$, so the quadratic remainder is at least $O(t^{2N+2})$; allowing coarse bounds and Laurent shifts gives the stated precision-doubling behavior. Every higher Taylor term is still smaller. $\square$

6.5 The Lagrange–Bürmann reversion formula

The most explicit formula requires no recursive composition.

Theorem 6.3 (Reversion at infinity). *Let $F = x + A$ with $A \in \mathcal{P}_{\geq 0}$. Its compositional inverse is*

$$F^{(-1)}(x) = x + \sum_{r \geq 1} \frac{(-1)^r}{r!} D^{r-1}(A(x)^r). \quad (6.8)$$

The sum is t -adically well-defined: its r -th term has t -order at least $r - 1$.

Proof. Introduce a marker λ and invert $F_\lambda(y) = y + \lambda A(y)$ as a formal series in λ . The one-variable Lagrange–Bürmann formula gives

$$F_\lambda^{(-1)}(x) = x + \sum_{r \geq 1} \frac{(-\lambda)^r}{r!} D^{r-1}(A(x)^r); \quad (6.9)$$

see, for example, [7]. In ordinary formal-power-series applications, setting $\lambda = 1$ need not be legitimate without a second filtration. Here it is: A^r has nonnegative t -order and each of the $r - 1$ derivatives raises t -order by at least one. Therefore only $r \leq N + 1$ can contribute modulo t^{N+1} , so the specialization $\lambda = 1$ is locally finite. Formula (6.9) composes with F_λ to x coefficientwise in λ , hence (6.8) is the inverse in the complete t -adic ring. $\square$

Remark 6.4. The decisive small quantity is not necessarily A itself. For $A = \log x$, one has $A \rightarrow \infty$, but $A/x = t\ell \rightarrow 0$, and each derivative in (6.8) supplies the needed inverse power of x . This is why reversion at infinity accommodates logarithmically large corrections.

6.6 Universal differential-polynomial form

Expanding the first terms of (6.8) gives

$$\begin{aligned} F^{(-1)}(x) &= x - A + AA' - A(A')^2 - \frac{1}{2}A^2A'' \\ &\quad + A(A')^3 + \frac{3}{2}A^2A'A'' + \frac{1}{6}A^3A''' + \dots \end{aligned} \quad (6.10)$$

This is organized by the Lagrange index r , not by inverse-power order. To obtain a fixed t -order, expand each differential polynomial and collect. Formula (6.10) is useful for hand derivations; (6.8) or Newton iteration is safer for automation.

6.7 Coefficient formula in the PL basis

Write

$$A = \sum_{m \geq 0} a_m(\ell) t^m, \quad A^r = \sum_{m \geq 0} A_{r,m}(\ell) t^m, \quad (6.11)$$

and

$$F^{(-1)}(x) = x + \sum_{N \geq 0} b_N(\ell) t^N.$$

Applying Theorem 4.2 termwise in (6.8) gives the finite formula

$$b_N(\ell) = \sum_{r=1}^{N+1} \frac{(-1)^r}{r!} \left[\prod_{s=N-r+1}^{N-1} (\partial_\ell - s) \right] A_{r, N-r+1}(\ell), \quad (6.12)$$

where an empty product is the identity. This formula reduces reversal to multiplication of A , polynomial differentiation, and a finite sum.

Using a dot for $d/d\ell$, the first three coefficients are

$$b_0 = -a_0, \quad (6.13)$$

$$b_1 = -a_1 + a_0 \dot{a}_0, \quad (6.14)$$

$$b_2 = -a_2 + a_0(\dot{a}_1 - a_1) + a_1 \dot{a}_0 - a_0(\dot{a}_0)^2 + \frac{1}{2} a_0^2(\dot{a}_0 - \ddot{a}_0). \quad (6.15)$$

They display two sources of terms: convolution among distinct inverse-power coefficients, and differentiation of their logarithmic dependence.

6.8 Exact check: a quadratic inverse and Catalan numbers

Take

$$F(y) = y + \frac{c}{y}.$$

The branch asymptotic to x is explicitly

$$F^{(-1)}(x) = \frac{x + \sqrt{x^2 - 4c}}{2} = x - \frac{c}{x} - \frac{c^2}{x^3} - \frac{2c^3}{x^5} - \frac{5c^4}{x^7} - \dots. \quad (6.16)$$

The coefficients are Catalan numbers. In (6.8), $A = c/x = ct$, and only odd inverse powers survive. This example simultaneously checks the signs, the derivative order $r - 1$, and the branch selection implicit in the near-identity condition.

6.9 A generic logarithmic-rational perturbation

Let

$$F(y) = y + a \log y + \frac{b}{y}. \quad (6.17)$$

Then, with $\ell = \log x$, its inverse begins

$$\begin{aligned} F^{(-1)}(x) &= x - a\ell + \frac{a^2\ell - b}{x} \\ &\quad + \frac{a^3(\ell^2/2 - \ell) + ab(1 - \ell)}{x^2} \\ &\quad + \frac{a^4(\ell^3/3 - 3\ell^2/2 + \ell) + a^2b(-\ell^2 + 3\ell - 1) - b^2}{x^3} + O(x^{-4}\ell^4). \end{aligned} \quad (6.18)$$

Setting $b = 0$ isolates the rescaled Lambert pattern; setting $a = 0$ recovers (6.16). The mixed terms quantify how a rational perturbation changes the Lambert polynomials.

6.10 Choosing an inversion algorithm

Method	Main advantage	Main limitation
Picard fixed point	shortest proof; monotone coefficient stabilization	roughly one new level per iteration; repeated work
Triangular matching	transparent hand calculation; exact next coefficient	sequential; weak use of fast multiplication
Newton	precision doubling; best high-order architecture	requires composition, derivative, and division at changing precision
Lagrange–Bürmann	closed formula; ideal for structural proofs and special families	powers A^r and high derivatives can be expensive if formed naively

A mature implementation usually combines them: obtain a few seed terms by triangular matching, use Newton for high precision, and use the Lagrange formula as an independent exact test.

7 Lambert W and the Lambert polynomials

7.1 From $we^w = z$ to a near-identity inverse

The Lambert function is defined by

$$W(z)e^{W(z)} = z. \quad (7.1)$$

It has branches W_k , $k \in \mathbb{Z}$. Taking a consistent logarithm converts (7.1) to

$$w + \log w = \text{Log } z + 2\pi i k. \quad (7.2)$$

Thus the large-argument problem is exactly the reversion problem for

$$f(w) = w + \log w. \quad (7.3)$$

The NIST Digital Library of Mathematical Functions uses this relation to state the all-branch asymptotic expansion [4]; the classical detailed analytic treatment is Corless et al. [3].

For a real variable $x \rightarrow +\infty$, define Wright's omega function by

$$\omega(x) + \log \omega(x) = x, \quad \omega(x) = W_0(e^x). \quad (7.4)$$

The outer function in (7.3) is $x + A(x)$ with $A = \ell$. Applying Theorem 6.3 immediately gives

$$\omega(x) = x + \sum_{r \geq 1} \frac{(-1)^r}{r!} D^{r-1} \ell^r. \quad (7.5)$$

Because the r -th summand begins at $x^{-(r-1)}$, exactly one Lagrange index contributes to each inverse-power level. This produces an especially clean polynomial family.

7.2 Definition and differential-operator formula

Definition 7.1 (Lambert polynomials). Define $L_n(X) \in \mathbb{Q}[X]$, $n \geq 0$, by

$$L_n(X) = \frac{(-1)^{n+1}}{(n+1)!} \left[\prod_{j=0}^{n-1} (\partial_X - j) \right] X^{n+1}, \quad (7.6)$$

with an empty product when $n = 0$. Thus $L_0(X) = -X$.

Then (7.5) is

$$\omega(x) = x + \sum_{n \geq 0} \frac{L_n(\ell)}{x^n}. \quad (7.7)$$

The degree of L_n is n for $n \geq 1$, its leading coefficient is $1/n$, and its linear coefficient is $(-1)^{n+1}$.

Derivation of (7.6). In (7.5), put $r = n + 1$. By Theorem 4.2 with initial t -level zero,

$$D^n \ell^{n+1} = t^n \prod_{j=0}^{n-1} (\partial_\ell - j) \ell^{n+1}.$$

The coefficient of $t^n = x^{-n}$ is precisely (7.6) after replacing ℓ by the polynomial variable X . $\square$

7.3 Integral recurrence

The same polynomials satisfy a remarkably simple recurrence, emphasized in Cohen's treatment [2].

Theorem 7.2 (Lambert recurrence). *With $L_0(X) = -X$,*

$$L_{n+1}(X) = -L_n(X) + n \int_0^X L_n(u) du, \quad n \geq 0. \quad (7.8)$$

Proof. Let

$$\mathcal{L}(T, X) = \sum_{n \geq 0} L_n(X) T^n. \quad (7.9)$$

Substitute $\omega(x) = x + \mathcal{L}(t, \ell)$ into $\omega + \log \omega = x$. Since

$$\log(x + \mathcal{L}(t, \ell)) = \ell + \log(1 + t \mathcal{L}(t, \ell)),$$

we obtain the implicit generating equation

$$\log(1 + T \mathcal{L}(T, X)) + \mathcal{L}(T, X) = -X. \quad (7.10)$$

Differentiating (7.10) with respect to T and simplifying gives

$$T \mathcal{L} \partial_T \mathcal{L} + (1 + T) \partial_T \mathcal{L} + \mathcal{L} = 0. \quad (7.11)$$

Equating coefficients, or equivalently differentiating the proposed recurrence, gives

$$L'_{n+1}(X) + L'_n(X) = n L_n(X), \quad L_{n+1}(0) = L_n(0) = 0,$$

which integrates to (7.8). The initial polynomial is fixed by the constant term of (7.10). $\square$

The recurrence computes L_{n+1} using one exact polynomial integral and one subtraction. It is usually the fastest way to generate a table over $\mathbb{Q}$. The differential-operator formula is more useful for proving identities.

7.4 Stirling-number formula

Let $s(n, k)$ denote the signed Stirling numbers of the first kind, characterized by

$$z^n = z(z-1)\cdots(z-n+1) = \sum_{k=0}^n s(n, k)z^k, \quad s(n+1, k) = s(n, k-1) - ns(n, k).$$

Let $[n]_k = (-1)^{n-k}s(n, k)$ be the unsigned Stirling cycle numbers.

Theorem 7.3 (Explicit Lambert polynomial). *For $n \geq 1$,*

$$L_n(X) = (-1)^{n+1} \sum_{m=1}^n s(n, n-m+1) \frac{X^m}{m!}, \quad (7.12)$$

$$= \sum_{m=1}^n (-1)^{n+m} \left[\begin{matrix} n \\ n-m+1 \end{matrix} \right] \frac{X^m}{m!}. \quad (7.13)$$

Proof. Substitute (7.12) into (7.8). The coefficient of $X^m/m!$ in $-L_n + n \int_0^X L_n$ becomes, after shifting the integral index, exactly the Stirling recurrence $s(n+1, k) = s(n, k-1) - ns(n, k)$. The initial case $L_1 = X$ is immediate. Equation (7.13) follows from the relation between signed and unsigned Stirling numbers. $\square$

Formula (7.13) is exactly the polynomial content of the standard DLMF all-branch expansion. It also shows that a common denominator of L_n divides $n!$; more precisely, the coefficient of X^m is an integer divided by $m!$.

7.5 Table of Lambert polynomials

Table 1: Lambert polynomials $L_n(X)$ through $n = 10$.

n	$L_n(X)$
0	$-X$
1	X
2	$\frac{X^2}{2} - X$
3	$\frac{X^3}{3} - \frac{3X^2}{2} + X$
4	$\frac{X^4}{4} - \frac{11X^3}{6} + 3X^2 - X$
5	$\frac{X^5}{5} - \frac{12X^4}{25} + \frac{35X^3}{6} - 5X^2 + X$
6	$\frac{X^6}{6} - \frac{137X^5}{137} + \frac{75X^4}{75} - \frac{85X^3}{245} + \frac{15X^2}{175} - X$
7	$\frac{X^7}{7} - \frac{20}{49X^6} + \frac{15}{203X^5} - \frac{8}{245X^4} + \frac{6}{175X^3} - \frac{21X^2}{2} + X$
8	$\frac{X^8}{8} - \frac{140}{363X^7} + \frac{180}{3283X^6} - \frac{120}{6769X^5} + \frac{3}{245X^4} - \frac{161X^3}{3} + 14X^2 - X$
9	$\frac{X^9}{9} - \frac{280}{761X^8} + \frac{1260}{29531X^7} - \frac{20}{1869X^6} + \frac{40}{7483X^5} - 189X^4 + 91X^3 - 18X^2 + X$
10	$\frac{X^{10}}{10} - \frac{280}{7129X^9} + \frac{1260}{6515X^8} - \frac{20}{9046X^7} + \frac{5985X^6}{16} - \frac{21091X^5}{40} + \frac{1575X^4}{4} - 145X^3 + \frac{45X^2}{2} - X$

The factored forms can be preferable numerically because every L_n , $n \geq 1$, contains the factor X . The expanded forms make comparison with Stirling numbers and coefficient recurrences immediate.

7.6 The complete branch expansion

Let

$$\xi_k = \text{Log } z + 2\pi i k, \quad (7.14)$$

with Log the chosen principal logarithm and with the branch of $\log \xi_k$ chosen consistently in the sector under consideration. Then

Theorem 7.4 (Lambert W expansion). *As $|z| \rightarrow \infty$ in a sector avoiding the relevant cuts,*

$$W_k(z) \sim \xi_k + \sum_{n \geq 0} \frac{L_n(\log \xi_k)}{\xi_k^n}. \quad (7.15)$$

Equivalently,

$$W_k(z) \sim \xi_k - \log \xi_k + \sum_{n \geq 1} \frac{(-1)^n}{\xi_k^n} \sum_{m=1}^n \left[\begin{matrix} n \\ n-m+1 \end{matrix} \right] \frac{(-\log \xi_k)^m}{m!}. \quad (7.16)$$

Proof. Equation (7.2) asks for the inverse of $w \mapsto w + \log w$ evaluated at ξ_k . Formula (7.7), interpreted with consistent complex logarithms, gives (7.15). Substitution of (7.13) gives (7.16). Formal substitution back into $w + \log w = \xi_k$ verifies every coefficient recursively. Analytic validity in sectors follows from the standard Lambert W asymptotic theory [3, 4]. $\square$

Writing $L_1 = \text{Log } z$ and $L_2 = \text{Log } L_1$ on the principal branch, (7.15) begins

$$\begin{aligned} W_0(z) \sim L_1 - L_2 + \frac{L_2}{L_1} + \frac{L_2(L_2 - 2)}{2L_1^2} \\ + \frac{L_2(2L_2^2 - 9L_2 + 6)}{6L_1^3} + \frac{L_2(3L_2^3 - 22L_2^2 + 36L_2 - 12)}{12L_1^4} + \dots \end{aligned} \quad (7.17)$$

This is a PL series in the large variable L_1 , with logarithm $L_2 = \log L_1$.

7.7 The real nonprincipal branch W_{-1}

Let $X \rightarrow +\infty$ and set $z = -e^{-X}$. Write

$$W_{-1}(-e^{-X}) = -u(X). \quad (7.18)$$

Then $u > 1$ and

$$u - \log u = X. \quad (7.19)$$

This is again a near-identity inverse, now with correction $-\log u$. Put $L = \log X$. Lagrange reversion gives

$$\begin{aligned} W_{-1}(-e^{-X}) = -X - L - \frac{L}{X} + \frac{L^2/2 - L}{X^2} \\ - \frac{L^3/3 - 3L^2/2 + L}{X^3} + \frac{L^4/4 - 11L^3/6 + 3L^2 - L}{X^4} + \dots \end{aligned} \quad (7.20)$$

The alternating pattern comes from replacing $A = +\log x$ by $A = -\log x$ before negating the inverse. Formula (7.20) is often numerically more convenient than forcing a complex branch formula onto a negative real argument.

7.8 Asymptotic versus convergent use

DLMF states (7.16) as a Poincaré asymptotic expansion. Cohen proves a stronger convergent representation. In his notation, for a nonzero parameter K and

$$M = K + \log K - \operatorname{Log} z - 2\pi i k,$$

one has

$$W_k(z) = K + \sum_{n \geq 0} \frac{L_n(M)}{K^n} \quad (7.21)$$

whenever, as a sufficient condition,

$$2(1 + |M|) < |K|. \quad (7.22)$$

Taking $K = \xi_k$ makes $M = \log \xi_k$ and recovers the canonical series (7.15). Thus the same formal object has three useful roles:

1. a coefficient identity in the PL ring;
2. a Poincaré expansion in broad large-argument sectors;
3. an actually convergent series under a sufficiently strong size condition.

One should not infer monotone error decrease from convergence: individual Lambert polynomials change sign, so neighboring truncations can occasionally have slightly different cancellation behavior.

7.9 Numerical illustration

Define the truncation

$$\omega_N(x) = x + \sum_{n=0}^N \frac{L_n(\log x)}{x^n}. \quad (7.23)$$

The following absolute errors were computed at high precision against $W_0(e^x)$.

x	$N = 0$	$N = 2$	$N = 4$	$N = 6$	$N = 10$
10	2.3201×10^{-1}	1.7370×10^{-3}	1.9026×10^{-5}	3.8679×10^{-8}	6.4406×10^{-10}
25	1.3180×10^{-1}	9.4510×10^{-5}	8.5143×10^{-7}	7.3947×10^{-9}	3.0268×10^{-13}
100	4.6657×10^{-2}	5.2671×10^{-6}	5.6023×10^{-9}	2.0334×10^{-12}	1.3940×10^{-18}

The table is not a proof of convergence, but it demonstrates the practical strength of exact log-polynomial coefficients. At $x = 100$, only eleven Lambert polynomials already give about eighteen correct decimal digits.

7.10 Derivative and quotient identities

Differentiating $W(z)e^{W(z)} = z$ gives

$$W'(z) = \frac{W(z)}{z(1 + W(z))}. \quad (7.24)$$

For $\omega(x) = W(e^x)$,

$$\omega'(x) = \frac{\omega(x)}{1 + \omega(x)} = 1 - \frac{1}{1 + \omega(x)}. \quad (7.25)$$

Once ω is known as a PL series, the right side is computed by the reciprocal recurrence of Section 3. Since $1 + \omega = x(1 + O(t\ell))$, its reciprocal lies in $\mathcal{P}$. This provides a strong independent check on the differentiated Lambert polynomial expansion.

Cohen derives, in the same notation as (7.21), the convergent identity

$$\frac{1}{1 + W_k(z)} = \sum_{n \geq 1} \frac{L'_n(M)}{K^n} \quad (7.26)$$

under the corresponding convergence conditions. The fact that L'_n is monic of degree $n - 1$ makes the leading behavior transparent.

7.11 Generalized Lambert equations

The equation

$$y + a \log y = x, \quad a \neq 0, \quad (7.27)$$

has the exact solution

$$y = a W_k \left(\frac{e^{x/a}}{a} \right), \quad (7.28)$$

with branch and logarithm choices matched to the desired solution. Indeed, with $u = y/a$, equation (7.27) becomes $u + \log u = x/a - \log a$. Its large solution is therefore a shifted and scaled omega expansion. The PL reversion machinery remains useful even when one does not recognize this closed form, and it extends immediately to additional terms such as b/y , as in (6.18).

More generally,

$$y^\alpha e^{\beta y} = z, \quad \alpha\beta \neq 0, \quad (7.29)$$

has solutions

$$y = \frac{\alpha}{\beta} W_k \left(\frac{\beta}{\alpha} z^{1/\alpha} \right). \quad (7.30)$$

Fractional powers and complex branches may enlarge the coefficient field, but the subsequent large-argument expansion is still polynomial-logarithmic in the appropriate leading logarithm.

8 Algorithms and exact implementation

8.1 A canonical in-memory representation

A truncated PL series should be stored as a finite map

$$n \mapsto p_n(\ell), \quad n_{\min} \leq n < N, \quad (8.1)$$

where each p_n is an exact polynomial over a chosen coefficient field. The exclusive cutoff N is part of the value's precision contract. A practical record contains:

- the least allowed exponent $n_{\min}$, which may be negative;
- the exclusive cutoff N ;
- a sparse map or dense array of coefficient polynomials;
- a coefficient-domain object supporting exact addition, multiplication, division by units, and differentiation in ℓ ;
- optional branch metadata for constants such as $\log c$.

Zero polynomials should be removed or normalized immediately. Otherwise a stale zero at the nominal lower exponent will corrupt order tests, unit tests, and Newton precision estimates.

Two storage strategies are common.

Dense in t , dense in ℓ . Best when nearly every inverse-power level is present and log degrees grow regularly, as for Lambert polynomials.

Sparse in t , sparse in ℓ . Best for differential equations or composed expressions with many structural zeros. Hash maps are convenient, but ordered vectors often have lower overhead and deterministic traversal.

The coefficient ring should be a generic parameter. The same series engine can then operate over $\mathbb{Q}[\ell]$, $\mathbb{Q}(a, b)[\ell]$, algebraic-number fields, or floating-point balls. The algebraic algorithms do not change.

8.2 Precision discipline

Every public operation should document whether the cutoff is absolute or relative. In this article,

$$\text{truncate}(F, N) = F \bmod t^N$$

uses an absolute exponent. Thus $x = t^{-1}$ is retained at every cutoff $N > -1$, and multiplying a series starting at t^{-3} by one starting at t^2 produces a result starting at t^{-1} .

A safe implementation follows three rules.

1. Never infer a missing coefficient to be unknown: within the advertised range it is exactly zero; outside the range it is unavailable.
2. Before an operation, determine which input range can influence the desired output range. Do not merely truncate both operands to the same number of stored entries.
3. In Newton algorithms, increase the absolute cutoff geometrically but retain all already known lower Laurent levels.

For compositions and derivatives, internal guard terms may be required. For example, to compute $D^k F \bmod t^N$, only coefficients of F below t^{N-k} are needed, while applying a primitive can lower the t -exponent by one.

8.3 Multiplication

In a coefficient-ring model, truncated multiplication uses exactly the convolution of [Theorem 3.1](#).

Listing 1: Naive truncated PL multiplication.

```
function Mul(F, G, cutoff N):
  R := zero series with cutoff N
  for each (i, p) in F.coefficients:
    for each (j, q) in G.coefficients:
      if i + j < N:
        R[i + j] := R[i + j] + p*q
  normalize R
  return R
```

The loops should also respect the known input cutoffs. If coefficients are dense, the inner range can be clipped analytically rather than tested on every iteration. Karatsuba, Toom–Cook, or FFT multiplication can be used at the outer t -level and again inside the coefficient polynomials, but for moderate orders the simple nested convolution is often faster and much easier to verify.

8.4 Unit inversion and division

Listing 2: Reciprocal of a PL unit starting at exponent zero.

```

function InvertUnit(G, cutoff N):
  require G.lower == 0
  c := G[0]
  require c is a nonzero scalar in K
  B[0] := 1/c
  for n := 1 to N-1:
    s := 0
    for k := 1 to n:
      s := s + G[k]*B[n-k]
    B[n] := -s/c
  return B

```

For a Laurent unit $G = t^m \tilde{G}$, shift the exponent, invert $\tilde{G}$, and shift back by $-m$. Quotient computation can either multiply by the reciprocal or use the direct recurrence (3.6). The direct recurrence saves one temporary series and usually performs fewer coefficient multiplications when only one numerator is involved.

If the leading coefficient is a polynomial rather than a scalar, the API should not silently coerce to rational functions. Prefer one of three explicit operations:

- `TryDivideInPL`, which checks exact polynomial divisibility at each step;
- `DivideInRationalLog`, which returns coefficients in $\mathbb{K}(\ell)$;
- `ExpandDivideAtLogInfinity`, which returns an ℓ^{-1} -truncated element of $\mathcal{H}$.

This makes representation growth visible to the caller.

8.5 Differentiation and shifted operators

Polynomial differentiation in ℓ is cheap. The PL derivative is:

Listing 3: Differentiation in the large variable x .

```

function Dx(F, cutoff N):
  R := zero series with cutoff N
  for each (n, p) in F.coefficients:
    if n + 1 < N:
      R[n + 1] := DLog(p) - n*p
  normalize R
  return R

```

For repeated derivatives of a single coefficient, applying $\Delta_{n,k}$ as a product of first-order operators is more stable than first expanding it as a degree- k polynomial in ∂_ℓ . Since the polynomial degree decreases under ∂_ℓ , many branches terminate early.

An implementation can cache

$$\Delta_{n,k}p = (\partial_\ell - n - k + 1)(\Delta_{n,k-1}p)$$

when the same outer coefficient participates in many Taylor terms.

8.6 Near-identity composition

A direct Taylor algorithm follows [Theorem 5.4](#).

Listing 4: Near-identity composition $F \circ (x + H)$.

```

function ComposeNear(F, H, cutoff N):
  # H has order >= 0; the inner series is x + H.
  R := 0
  Hpow := 1
  DkF := truncate(F, N)
  fact := 1
  k := 0

  while order(DkF) < N:
    R := truncate(R + Hpow*DkF/fact, N)
    k := k + 1
    Hpow := truncate(Hpow*H, N)
    DkF := Dx(DkF, N)
    fact := fact*k
  return R

```

This transparent algorithm is excellent at low and medium orders. Several improvements are possible:

1. Generate $H^k/k!$ recursively by multiplication by H/k .
2. Stop each polynomial operation at the degree bound implied by the desired output level.
3. Group Taylor terms in blocks and use baby-step/giant-step evaluation for high precision.
4. For H with positive t -order $h > 0$, use the sharper bound $\text{ord}_{x^{-1}}(H^k D^k F) \geq \text{ord}_{x^{-1}} F + k(h + 1)$.

For a general admissible inner series $G = cx^q(1 + U)$, the implementation of (5.3) needs three reusable primitives:

$$\begin{aligned}
 V &= \log(1 + U), \\
 B_n &= (1 + U)^{-n}, \\
 p_n(q\ell + \log c + V) &= \sum_j p_{n,j}(q\ell + \log c + V)^j.
 \end{aligned}$$

The powers B_n can be updated by multiplying by $(1 + U)^{-1}$, and powers of $q\ell + \log c + V$ can be shared among all p_n up to the maximum needed log degree.

8.7 Newton reversion

Represent the unknown inverse as $G = x + B$. A precision-doubling loop is:

Listing 5: Newton series reversal in the PL ring.

```

function ReverseNearIdentity(F, target cutoff N):
  require F = x + A with order(A) >= 0
  G := x                # correct at leading level
  p := 0                # correction known below t^p

  while p < N:
    q := min(N, max(1, 2*p + 2))
    R := truncate(Compose(F, G, q) - x, q)
    J := truncate(Compose(Dx(F, q), G, q), q)
    C := truncate(Divide(R, J, q), q)
    G := truncate(G - C, q)
    p := q
  return truncate(G, N)

```

The exact initialization convention for p depends on whether the constant correction is counted as level zero. The invariant that matters is verified, not assumed:

$$F \circ G - x \equiv 0 \pmod{t^p}.$$

After each step, recompute the residual order in debug builds. This catches nearly every indexing error in derivative shifts and cutoffs.

One may use the inverse derivative identity

$$(F^{\langle -1 \rangle})' = \frac{1}{F' \circ F^{\langle -1 \rangle}} \quad (8.2)$$

as an additional consistency check.

8.8 Lambert-polynomial generation

The recurrence (7.8) is particularly simple.

Listing 6: Exact Lambert polynomials.

```
function LambertPolynomials(N):
  L[0] := -X
  for n := 0 to N-1:
    L[n+1] := -L[n] + n*IntegralFromZero(L[n], X)
    normalize L[n+1]
  return L
```

The table should be cross-checked against either (7.6) or (7.12). With rational coefficients and dense polynomials, producing $L_0, \dots, L_N$ takes $O(N^2)$ scalar arithmetic operations, apart from integer/rational bit growth.

8.9 Complexity in two models

Let N be the number of retained inverse-power levels and let M bound the log degree. There are two useful cost models.

Coefficient-ring model. Treat addition, multiplication, division, and differentiation in $\mathbb{K}[\ell]$ as unit-cost operations. Naive series multiplication and unit inversion require $O(N^2)$ coefficient operations. Differentiation is $O(N)$. Straight Taylor composition is typically $O(N^3)$ if every power and product is formed naively, though degree/order bounds often reduce the constant substantially.

Scalar model. Dense multiplication of two degree- M polynomials costs $O(M^2)$ scalar operations naively. Hence naive PL multiplication costs $O(N^2 M^2)$ in a uniform-degree bound. Using the actual profile $d_F(n)$ gives a much sharper sum

$$\sum_{r < N} \sum_{i+j=r} O((d_F(i) + 1)(d_G(j) + 1)).$$

Fast univariate series and polynomial multiplication can lower these bounds. Fast composition is more subtle and depends on the coefficient ring and memory model. The important architectural point is that Newton reversion inherits the cost of composition and division at geometrically increasing cutoffs; it does not add another factor proportional to the final precision when implemented with precision doubling.

8.10 Verification strategy

A symbolic transseries engine should be tested by identities, not by printed coefficient snapshots alone. Recommended exact tests include:

1. $T_{<N}(F(GH)) = T_{<N}((FG)H)$ for randomly generated truncated series.
2. $T_{<N}(G G^{-1}) = 1$ for units with random polynomial coefficients.
3. The product, quotient, and chain rules.
4. $F \circ F^{(-1)} = x$ and $F^{(-1)} \circ F = x$ independently.
5. Equality of Picard, triangular, Newton, and Lagrange inverses through a common cutoff.
6. Equality of the recurrence, differential-operator, and Stirling formulas for every generated Lambert polynomial.
7. Substitution of a truncated Lambert expansion into $w + \log w - x$, whose residual must begin exactly beyond the requested order.

For numerical tests, evaluate at several large x , but never substitute a single floating-point test for a formal identity. Catastrophic cancellation can make an algebraically wrong coefficient appear accurate at one point.

9 Analytic meaning and remainder control

9.1 A level-wise definition

Let f be a function on a positive ray. We write

$$f(x) \sim \sum_{n \geq n_0} p_n(\log x) x^{-n} \quad (x \rightarrow +\infty) \quad (9.1)$$

when for every integer $N \geq n_0$,

$$f(x) - \sum_{n=n_0}^N p_n(\log x) x^{-n} = o(x^{-N}). \quad (9.2)$$

Because any nonzero polynomial in $\log x$ fails to tend to zero, condition (9.2) determines each whole coefficient polynomial uniquely. It also permits the next omitted level to contain any fixed power of $\log x$, since

$$x^{-N-1}(\log x)^d = o(x^{-N})$$

for every fixed d .

A stronger quantified form is

$$R_N(x) = O(x^{-N-1}(1 + \log x)^{m_N}) \quad (9.3)$$

for some degree bound m_N . Such bounds are useful for proving closure under composition and for uniform numerical error estimates.

On a complex sector, replace $\log x$ by a fixed analytic branch and require the estimate uniformly on every closed subsector. Branch selection is part of the expansion data.

9.2 Algebraic operations on analytic expansions

Suppose f and g have PL expansions with remainders satisfying bounds of the form (9.3). Then addition and multiplication can be performed formally: multiplying remainders by retained log-polynomial terms preserves the inverse-power hierarchy. Division is valid when the leading analytic factor stays away from zero and its formal leading coefficient is a unit in the chosen coefficient field.

For example, if

$$g(x) = cx^q(1 + u(x)), \quad c \neq 0, \quad u(x) = O(x^{-1}(\log x)^d),$$

then $1/g$, $\log g$, and g^α admit the expansions obtained from the geometric, logarithmic, and binomial series. Uniform smallness of u on the domain justifies the analytic operations; local finiteness justifies the formal coefficient calculations.

9.3 Differentiation is not automatic

A formal PL series can always be differentiated, but an asymptotic expansion cannot always be differentiated termwise without hypotheses on the remainder. There exist smooth functions that are smaller than every power of x^{-1} while their derivatives are large on sparse sets. Therefore

$$f \sim F \not\Rightarrow f' \sim F'$$

in complete generality.

Sufficient conditions include:

- holomorphy in a complex sector with uniform remainder estimates, allowing Cauchy estimates;
- membership in a symbol class with derivative bounds $D^j R_N = O(x^{-N-1-j}(\log x)^{m_{N,j}})$;
- a Hardy-field or comparable tame-germ setting in which differentiation is compatible with asymptotic comparison.

The distinction is one reason general transseries are studied as differential fields of formal objects separately from their realizations as germs [10, 1].

9.4 Composition of analytic expansions

Let $f(y)$ have a PL expansion for y in a sector, and let

$$g(x) = cx^q(1 + u(x))$$

map the working x -sector into that y -sector for large $|x|$. If u has a uniform PL expansion and the remainders of f are uniform after substitution, then the formal composition theorem predicts the analytic expansion of $f(g(x))$. Two checks are indispensable:

1. $g(x)$ must not cross a branch cut of any logarithm used by f ;
2. the omitted remainder of f must remain smaller than the desired output order after replacing y by $g(x)$.

For $q > 0$, powers of g^{-1} preserve order in a controlled way, which is the analytic counterpart of the finite-contribution proof in Theorem 5.1.

9.5 Analytic inversion

The formal inverse of $F(x) = x + A(x)$ predicts an actual inverse when, for large real x , the function is continuous and strictly monotone with

$$F'(x) = 1 + o(1).$$

Then F' is eventually nonzero, and the inverse function theorem applies on a tail. Substitution of successive formal truncations gives residuals of increasing order; division by F' transfers those residual estimates to the inverse. On complex sectors, one uses eventual univalence or a suitable local inverse theorem.

The formal group theorem alone does not choose among globally distinct inverse branches. The condition $G(x) \sim x$ selects the branch near infinity. For Lambert W , the integer k and the logarithm branches in ξ_k carry the additional branch data.

9.6 Beyond-all-orders ambiguity

A PL expansion cannot detect exponentially small differences. For example,

$$f(x) \quad \text{and} \quad f(x) + e^{-x}$$

have the same PL expansion to every inverse-power order. Composition and reversion can transport such invisible terms, and in nonlinear problems they may encode Stokes jumps or free transseries parameters. The full logarithmic-exponential or resurgent transseries framework adjoins the exponential monomials needed to retain that information.

This limitation does not weaken the algebra developed here. It states exactly what information the PL quotient scale remembers: the complete power-log sector and nothing beyond it.

9.7 Convergent PL series

A formal series

$$\sum_{n \geq 0} p_n(\ell) t^n$$

may converge for a fixed numerical x even though the coefficient degree grows with n . A simple sufficient test is a bound

$$|p_n(\log x)| \leq C(x) R(x)^n$$

with $R(x) < |x|$. In Lambert's case, Cohen's construction supplies a concrete condition, (7.22), rather than relying on a rough coefficient estimate.

Convergence in t for each fixed ℓ should not be confused with uniform convergence as $x \rightarrow \infty$, because t and ℓ are linked by $t = e^{-\ell}$. The formal PL topology deliberately treats them as an ordered asymptotic pair, not as two independent small complex variables.

10 Boundary of the class and systematic enlargements

10.1 The closure decision begins with the leading factor

Every nonzero $F \in \mathcal{P}$ has a multiplicative decomposition

$$F = t^m p(\ell)(1 + U), \quad p \in \mathbb{K}[\ell] \setminus \{0\}, \quad U \in t\mathbb{K}(\ell)[[t]]. \quad (10.1)$$

The three factors diagnose most closure questions.

- The monomial $t^m = x^{-m}$ controls the primary power scale.
- The polynomial $p(\ell)$ controls whether division or a logarithm needs inverse powers of ℓ or $\log \ell$.
- The unit $1 + U$ is handled by local power-series calculus.

For example,

$$\log F = -m\ell + \log p(\ell) + \log(1 + U). \quad (10.2)$$

If p is a nonzero constant, $\log F \in \mathcal{P}$ after choosing that constant's logarithm. If p has degree $d > 0$, write

$$p(\ell) = a\ell^d \left(1 + \frac{c_1}{\ell} + \cdots + \frac{c_d}{\ell^d}\right).$$

Then

$$\log p(\ell) = \log a + d \log \ell + \log \left(1 + \frac{c_1}{\ell} + \cdots + \frac{c_d}{\ell^d}\right). \quad (10.3)$$

The last term is a Laurent series in ℓ^{-1} , while $d \log \ell$ creates one new iterated logarithm. Formula (10.3) pinpoints why $\mathcal{P}$ is not closed under logarithms of arbitrary nonzero elements.

10.2 Puiseux-log series

For a fixed positive integer s , define

$$\mathcal{P}^{(1/s)} = \mathbb{K}[\ell]((x^{-1/s})). \quad (10.4)$$

It is the minimal extension needed for rational leading powers. The derivative formula becomes

$$D \left(x^{-r/s} p(\ell) \right) = x^{-r/s-1} \left(p'(\ell) - \frac{r}{s} p(\ell) \right), \quad (10.5)$$

so differential closure is retained. Composition with $G = cx^{q/s}(1 + U)$ maps integer-power PL series into a common Puiseux-log extension determined by the denominators of all resulting exponents.

Puiseux behavior is also forced by critical inversion. If the leading linear derivative vanishes, Newton's unit condition fails. For instance, $y^2 = x$ has inverse $x^{1/2}$, and the Lambert W function near its branch point $-e^{-1}$ has a square-root local parameter. Such expansions are not a failure of reversion; they belong to a ramified local scale rather than the near-identity group $\mathcal{G}$.

10.3 Generalized powers and Hahn exponents

Instead of fixing a denominator, let Γ be an ordered additive subgroup of $\mathbb{R}$ or $\mathbb{C}$ equipped with an asymptotically meaningful real part. One may form well-based sums

$$\sum_{\gamma \in S} p_\gamma(\ell) x^{-\gamma}, \quad (10.6)$$

where $S \subseteq \Gamma$ is well ordered in the direction of increasing smallness. This is a logarithmic Hahn-series extension. The product is again convolution in Γ , and differentiation sends exponent γ to $\gamma + 1$. General transseries replace the single exponent group by recursively generated monomial groups, but the local-finiteness principle is the same [5, 10].

10.4 Laurent-log and rational-log coefficients

The fields $\mathcal{R}$ and $\mathcal{H}$ from Section 3.6 solve different problems.

$\mathcal{R} = \mathbb{K}(\ell)((t))$. Preserves exact rational dependence on ℓ . It is ideal for symbolic quotients and compact expressions, but it does not expand the secondary asymptotic scale.

$\mathcal{H} = \mathbb{K}((\ell^{-1}))((t))$. Resolves every rational-log coefficient into powers of ℓ^{-1} . It is better for asymptotic comparison and for composing with quantities where $\ell \rightarrow \infty$, but requires a second truncation policy.

An $\mathcal{H}$ implementation needs a rectangular or staircase precision region in the pair (n, j) . A finite t -cutoff alone does not bound the infinitely many negative powers of ℓ at one level. One possible contract is

$$n < N, \quad j > J(n),$$

with a level-dependent lower log exponent $J(n)$. The boundary should be chosen to be stable under the intended convolutions.

Example 10.1. Division by a nonunit illustrates the distinction:

$$\begin{aligned} \frac{1+t}{\ell+t} &= \frac{1+t}{\ell} \left(1 + \frac{t}{\ell}\right)^{-1} \\ &= \ell^{-1} + \sum_{n \geq 1} (-1)^{n-1} (\ell^{-n} - \ell^{-n-1}) t^n. \end{aligned} \quad (10.7)$$

The compact quotient belongs to $\mathcal{R}$; its secondary asymptotic expansion belongs to $\mathcal{H}$; neither belongs to $\mathcal{P}$.

10.5 Iterated logarithms

Set

$$\ell_1 = \log x, \quad \ell_2 = \log \ell_1, \quad \ell_3 = \log \ell_2, \quad \dots$$

The derivative hierarchy is

$$D\ell_1 = x^{-1}, \quad D\ell_2 = (x\ell_1)^{-1}, \quad D\ell_3 = (x\ell_1\ell_2)^{-1}, \quad \dots \quad (10.8)$$

A finite-depth logarithmic transseries ring allows polynomial or Laurent coefficients in successive ℓ_j , ordered lexicographically by

$$x \succ \ell_1 \succ \ell_2 \succ \dots \succ 1.$$

This enlargement becomes necessary as soon as a leading coefficient contains a positive power of ℓ_1 inside a logarithm. For instance,

$$\log(x\ell_1^a) = \ell_1 + a\ell_2.$$

It also arises in inverses of functions such as

$$y + \log y + \log \log y.$$

The leading inverse is $x - \log x - \log \log x + \dots$, so no one-log coefficient ring can be closed under the calculation.

At each finite depth, the same architecture survives: a well-ordered monomial scale, a filtered completion, unit recurrences, shifted differential operators, and near-identity reversion. What changes is the coefficient algebra and the bookkeeping for its own subordinate scales.

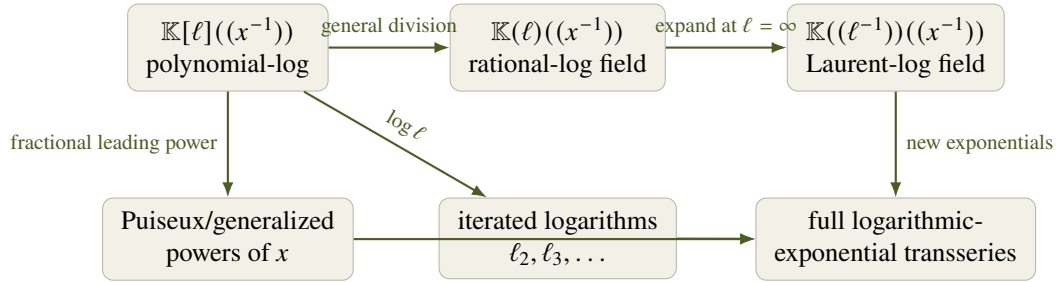
10.6 Exponentials and full logarithmic-exponential transseries

For $F \in \mathcal{P}$, write its additive decomposition according to size:

$$F = F_{\text{large}} + c + F_{\text{small}}.$$

If F_{large} is merely an integral multiple of ℓ , then e^F is a power of x times a small exponential and can remain in a power-log extension. If F_{large} contains x , ℓ^2 , or more general large terms, e^F creates a genuinely new monomial such as e^x or e^{ℓ^2} . Repeated closure under exponentials and logarithms leads to logarithmic-exponential transseries.

These larger fields are ordered differential fields with composition on a suitable positive-large domain and compositional inverses for broad classes [5, 6, 10]. The PL ring should be viewed not as a rival definition but as a computationally tractable convex fragment encountered after a dominant logarithmic change of variables.



10.7 A formal implicit-function principle

Series reversal is one instance of a more general Newton–Hensel mechanism. Let $\Phi(x, y)$ be an expression built from PL coefficients and local analytic operations in y . Suppose an approximation $Y_0 \in \mathcal{P}$ satisfies

$$\Phi(x, Y_0) \in t^N \mathcal{P}_{\geq 0}$$

and the partial derivative

$$\Phi_y(x, Y_0) = c + O(t), \quad c \in \mathbb{K}^\times,$$

is a unit. Then the Newton iteration

$$Y_{j+1} = Y_j - \frac{\Phi(x, Y_j)}{\Phi_y(x, Y_j)} \quad (10.9)$$

constructs a unique solution near Y_0 , with precision doubling under the same Taylor estimates as in Section 6. The leading-unit condition is the formal analogue of a nonzero Jacobian.

This principle covers algebraic equations, logarithmic implicit equations, Riccati-type algebraic balances, and parameter-dependent inversions. When Φ_y is not a unit, the correct response is usually not to force division, but to refine the dominant balance and enlarge the monomial scale.

11 Worked calculations

11.1 Logarithm of a monomial-leading PL series

Consider

$$F(x) = x^2 \left(1 + \ell t + (\ell^2 - 1)t^2 \right). \quad (11.1)$$

The leading coefficient after factoring x^2 is the scalar 1, so the logarithm stays in $\mathcal{P}$:

$$\log F = 2\ell + \log(1 + U), \quad U = \ell t + (\ell^2 - 1)t^2.$$

Expanding $U - U^2/2 + U^3/3 - U^4/4 + \dots$ gives

$$\log F = 2\ell + \ell t + \left(\frac{\ell^2}{2} - 1 \right) t^2 + \left(-\frac{2\ell^3}{3} + \ell \right) t^3$$

$$+ \left(\frac{\ell^4}{4} - \frac{1}{2} \right) t^4 + O(t^5). \quad (11.2)$$

Every coefficient at order t^N uses only powers U^r with $r \leq N$.

If the parenthesis in (11.1) were replaced by $\ell + 1 + O(t)$, the logarithm would begin with $\log \ell$ and leave the one-log ring. The difference is entirely in the leading coefficient.

11.2 Triangular construction of $\omega(x)$

Let $f(y) = y + \log y$. Starting with $G_0 = x$, compute the residual $R_j = f(G_j) - x$ and negate its leading coefficient as in (6.6):

$$\begin{aligned} G_0 &= x, & R_0 &= \ell, \\ G_1 &= x - \ell, & R_1 &= -\ell t - \frac{\ell^2}{2} t^2 - \frac{\ell^3}{3} t^3 - \dots, \\ G_2 &= x - \ell + \ell t, & R_2 &= -\left(\frac{\ell^2}{2} - \ell \right) t^2 + \left(-\frac{\ell^3}{3} + \ell^2 \right) t^3 + \dots, \\ G_3 &= x - \ell + \ell t + \left(\frac{\ell^2}{2} - \ell \right) t^2, & R_3 &= -\left(\frac{\ell^3}{3} - \frac{3\ell^2}{2} + \ell \right) t^3 + \dots. \end{aligned}$$

The next correction is therefore $L_3(\ell)t^3$. In general, if

$$G_N = x + \sum_{n=0}^{N-1} L_n(\ell) t^n,$$

then

$$f(G_N) - x = -L_N(\ell) t^N + O(t^{N+1} \ell^{N+1}). \quad (11.3)$$

This identity offers another characterization of the Lambert polynomials and a simple residual test for generated tables.

11.3 Inverting $y + a \log y + b/y + c/y^2$

Consider

$$x = y + a \log y + \frac{b}{y} + \frac{c}{y^2}. \quad (11.4)$$

Set $A(x) = a\ell + bt + ct^2$ in the Lagrange formula. The inverse branch asymptotic to x is

$$\begin{aligned} y &= x - a\ell + \frac{a^2\ell - b}{x} \\ &\quad + \frac{a^3(\ell^2 - 2\ell) - 2ab\ell + 2ab - 2c}{2x^2} \\ &\quad + \frac{1}{6x^3} \left(2a^4\ell^3 - 9a^4\ell^2 + 6a^4\ell - 6a^2b\ell^2 + 18a^2b\ell - 6a^2b \right. \\ &\quad \left. - 12ac\ell + 6ac - 6b^2 \right) + O(x^{-4}\ell^4). \end{aligned} \quad (11.5)$$

Several structural checks are visible:

- the highest log power at level x^{-n} is supplied solely by the repeated logarithmic perturbation;
- c , which begins at x^{-2} , first affects the inverse at the same level;
- products such as b^2 first occur when two rational perturbations can combine at a common level;
- setting $b = c = 0$ leaves scaled Lambert polynomials.

Substitution of (11.5) into (11.4) cancels every term through x^{-3} .

11.4 A quotient followed by composition

Let

$$Q(x) = \frac{1 + \ell t}{1 - \ell t + t^2}.$$

Since the denominator is a unit, recurrence (3.6) gives

$$Q = 1 + 2\ell t + (2\ell^2 - 1)t^2 + (2\ell^3 - 3\ell)t^3 + (2\ell^4 - 5\ell^2 + 1)t^4 + \dots. \quad (11.6)$$

Now compose with $G = x + \ell$. The Taylor formula says

$$Q \circ G = Q + \ell Q' + \frac{\ell^2}{2} Q'' + \dots.$$

To obtain terms through t^3 , only derivatives through order three can contribute, and the derivative of a coefficient $p_n(\ell)t^n$ is obtained by $(\partial_\ell - n)p_n$ at the next level. The calculation is finite at every stage; no direct expansion of a nested rational-log expression is required.

This example illustrates a useful workflow: normalize divisions first, cache the resulting coefficient polynomials, then compose. Reversing the order can force larger intermediate expressions even though the final truncated answer is the same.

11.5 Power-log inversion versus branch-point inversion

Compare two inverse problems.

$$\begin{aligned} x &= y + \log y, \\ x &= y^2 + y. \end{aligned}$$

The first has derivative $1 + 1/y \rightarrow 1$ and belongs to the near-identity PL group after dominant balance $y \sim x$. The second has dominant balance $y \sim x^{1/2}$, so the correct scale is Puiseux from the beginning:

$$y = \frac{-1 + \sqrt{1 + 4x}}{2} = x^{1/2} - \frac{1}{2} + \frac{1}{8}x^{-1/2} - \frac{1}{128}x^{-3/2} + \dots.$$

Trying an integer-power PL ansatz for the second equation cannot succeed. A dominant-balance analysis must precede coefficient recursion.

11.6 A practical closure checklist

Before performing a requested operation, ask:

1. What is the leading x -power, and is its exponent in the current exponent group?
2. Is the leading coefficient in ℓ a unit of the coefficient ring?
3. Does a logarithm act on a scalar-leading unit, or will it create $\log \ell$?
4. For composition, does the inner argument tend to infinity with a controlled monomial leading term?
5. For reversal, is the leading derivative a unit? If not, what ramified balance replaces the near-identity ansatz?
6. Which branch constants must be carried explicitly?
7. What output monomials are retained, and is that truncation region stable under all intermediate products?

These questions turn most apparent “transseries mysteries” into routine representation decisions.

12 Conclusions and a compact structural picture

Polynomial-logarithmic transseries occupy a useful middle ground. Ordinary Laurent series are too small to survive the inversion of even the elementary map $y \mapsto y + \log y$, while the full logarithmic-exponential transseries field contains much more structure than is needed for that problem. The ring

$$\mathcal{P} = \mathbb{K}[\ell]((t)), \quad t = x^{-1}, \quad \ell = \log x,$$

is therefore not merely a pedagogical toy. It is a sharply delimited computational domain in which support, truncation, and coefficient arithmetic are completely explicit.

The main conclusions of the preceding sections can be compressed as follows.

1. **Multiplication is unconditional.** Products are Cauchy convolutions over $\mathbb{K}[\ell]$, and every requested coefficient is a finite sum. The t -adic filtration makes truncated multiplication exact below the cutoff.
2. **Division is conditional.** A series is a unit of $\mathbb{K}[\ell]((t))$ precisely when its leading coefficient is a nonzero scalar. A nonconstant leading polynomial in ℓ forces rational-log coefficients; an inverse power of ℓ suggests Laurent-log or ℓ^{-1} -adic coefficients instead. This is a representation issue, not a failure of formal algebra.
3. **Differentiation and antidifferentiation are internal.** The identity

$$D(t^n p(\ell)) = t^{n+1}(\partial_\ell - n)p(\ell)$$

turns analytic differentiation into a shifted polynomial operator. The corresponding first-order polynomial equations also give an antiderivative within the same class.

4. **Composition is an admissible partial operation.** Substitution becomes locally finite when the inner series has a controlled large leading monomial. The most important case is the near-identity map $G = x + H$, for which Taylor expansion is t -adically finite because every derivative raises the inverse-power level. Arbitrary formal substitution is neither defined nor closed.
5. **Near-identity maps form a composition group.** Every $F = x + A$, with A of nonnegative t -order, has a unique inverse of the same form. Picard iteration proves existence, triangular matching exposes coefficient dependence, Newton iteration gives precision doubling, and Lagrange–Bürmann gives the closed formula

$$F^{\langle -1 \rangle} = x + \sum_{r \geq 1} \frac{(-1)^r}{r!} D^{r-1}(A^r).$$

6. **Dominant balance comes before algebra.** If the leading map behaves like cy^q , then its inverse behaves like $c^{-1/q}x^{1/q}$. A nonintegral exponent requires a Puiseux-log scale; vanishing or singular leading derivatives may require a different scale again. Reversion does not repair an incorrectly chosen monomial group.
7. **Lambert W is the canonical model.** Taking logarithms of $we^w = z$ reduces every large branch to inversion of $w + \log w$. The resulting Lambert polynomials simultaneously encode the coefficient recurrence, the Lagrange differential formula, and the signed Stirling numbers of the first kind.

The operational rule

Normalize the dominant monomial first. Then ask whether the leading coefficient is a unit, whether every logarithm is applied to a scalar-leading unit, and whether the inner argument of every

composition is large in a controlled way. Once those three questions have affirmative answers, coefficient extraction is finite and algorithmic. If one answer is negative, enlarge the scale before calculating.

The distinction among formal existence, analytic asymptotic validity, and convergence remains essential. Formal PL algebra determines coefficients and identities. Analytic theorems justify substitution into actual functions and control remainders. Convergence, when available, is an additional property, not part of the definition. Keeping these layers separate is what makes the small calculus reliable and extensible.

A Coefficient extraction in closed form

This appendix records formulas useful for direct implementations and for checking hand calculations. They contain no infinite sums at a fixed output level.

A.1 Finite convolution polynomials

Let

$$H = \sum_{j \geq 0} h_j(\ell) t^j.$$

For integers $m, r \geq 0$, define

$$C_{m,r}(h_0, h_1, \dots) = \sum_{\substack{j_1, \dots, j_r \geq 0 \\ j_1 + \dots + j_r = m}} h_{j_1} \cdots h_{j_r}, \quad (\text{A.1})$$

with $C_{0,0} = 1$ and $C_{m,0} = 0$ for $m > 0$. Equivalently,

$$H^r = \sum_{m \geq 0} C_{m,r} t^m.$$

The recurrence

$$C_{m,r+1} = \sum_{j=0}^m h_j C_{m-j,r} \quad (\text{A.2})$$

computes all values in a triangular table. If $h_0 = 0$, then $C_{m,r} = 0$ for $m < r$, and these are the ordinary partial convolution polynomials underlying ordinary Bell polynomials.

A.2 A coefficient formula for near-identity composition

For $n \in \mathbb{Z}$ and $r \geq 0$, set

$$\Delta_{n,r} = \prod_{q=0}^{r-1} (\partial_\ell - n - q), \quad \Delta_{n,0} = 1. \quad (\text{A.3})$$

Let

$$F = \sum_{n \geq n_0} f_n(\ell) t^n, \quad G = x + H, \quad H = \sum_{j \geq 0} h_j(\ell) t^j.$$

Then $F \circ G = \sum_N c_N(\ell) t^N$, where

$$c_N = \sum_{n=n_0}^N \sum_{r=0}^{N-n} \frac{C_{N-n-r,r}}{r!} \Delta_{n,r} f_n. \quad (\text{A.4})$$

All polynomial arguments $h_0, h_1, \dots$ in $C_{m,r}$ are suppressed in (A.4).

Proof. Taylor's formula and the repeated-derivative identity give

$$F(x + H) = \sum_{r \geq 0} \frac{H^r}{r!} D^r F = \sum_{r \geq 0} \sum_{m \geq 0} \sum_{n \geq n_0} \frac{C_{m,r}}{r!} \Delta_{n,r} f_n t^{m+n+r}.$$

The coefficient of t^N has $m = N - n - r$, which yields (A.4). The inequalities $m \geq 0$ and $r \geq 0$ make both displayed sums finite. $\square$

Two special cases are especially useful. If $H = h_0(\ell)$ has no positive t -levels, then only $m = 0$ occurs and

$$c_N = \sum_{n=n_0}^N \frac{h_0^{N-n}}{(N-n)!} \Delta_{n,N-n} f_n. \quad (\text{A.5})$$

Thus composition by $x + \ell$, $x + a\ell + b$, or any polynomial-logarithmic shift at level zero has a single-sum coefficient formula. If instead $h_0 = 0$, then the extra vanishing $C_{m,r} = 0$ for $m < r$ sharply reduces the summation region.

A.3 Faà di Bruno inside the PL ring

Let Φ be analytic at the scalar leading value of F , so that $\Phi(F)$ is defined by the local functional calculus of Section 3.6. For $m \geq 1$, the usual Faà di Bruno identity remains valid:

$$D^m \Phi(F) = \sum_{k=1}^m \Phi^{(k)}(F) B_{m,k}(F', F'', \dots, F^{(m-k+1)}), \quad (\text{A.6})$$

where the exponential partial Bell polynomial is

$$B_{m,k}(z_1, z_2, \dots) = \sum \frac{m!}{j_1! j_2! \dots} \prod_{r \geq 1} \left(\frac{z_r}{r!} \right)^{j_r}. \quad (\text{A.7})$$

The sum ranges over $\sum j_r = k$ and $\sum r j_r = m$. It is finite before any transseries substitution is made. Since differentiation, multiplication, and the relevant local analytic functions are all defined in the PL ring, the classical formal proof applies verbatim.

Formula (A.6) is useful in two places. First, it expands high Taylor derivatives occurring in composition. Second, it expresses the Lagrange terms $D^{r-1}(A^r)$ as differential polynomials in A , making universal inverse formulas independent of any chosen coefficient basis.

A.4 Fast recurrences for elementary local functions

The following recurrences use an *auxiliary* derivation ∂_t that holds ℓ fixed. It is legitimate for coefficient extraction in the abstract power-series ring $\mathbb{K}[\ell][[t]]$, but it is not the analytic derivative with respect to t , because analytically $\ell = -\log t$.

Let $U = \sum_{n \geq 1} u_n t^n$.

For $E = \exp U = \sum_{n \geq 0} e_n t^n$,

$$e_0 = 1, \quad n e_n = \sum_{j=1}^n j u_j e_{n-j}. \quad (\text{A.8})$$

This follows from $\partial_t E = E \partial_t U$.

For $L = \log(1 + U) = \sum_{n \geq 1} \lambda_n t^n$,

$$\lambda_n = u_n - \frac{1}{n} \sum_{j=1}^{n-1} (n-j) u_j \lambda_{n-j}. \quad (\text{A.9})$$

Indeed, $(1 + U)\partial_t L = \partial_t U$.

For $P = (1 + U)^\alpha = \sum_{n \geq 0} p_n t^n$,

$$p_0 = 1, \quad np_n = \alpha n u_n + \sum_{j=1}^{n-1} (\alpha j - (n-j)) u_j p_{n-j}. \quad (\text{A.10})$$

This comes from $(1 + U)\partial_t P = \alpha P \partial_t U$. The formulas are valid over any characteristic-zero coefficient algebra, hence in particular over $\mathbb{K}[\ell]$, $\mathbb{K}(\ell)$, or a Puiseux-log coefficient ring.

B Further details on reversal

B.1 Universal terms through the fifth Lagrange index

Grouping the inverse by Lagrange index rather than by t -level gives

$$\begin{aligned} F^{(-1)} &= x - A + AA' \\ &\quad - A(A')^2 - \frac{1}{2}A^2A'' \\ &\quad + A(A')^3 + \frac{3}{2}A^2A'A'' + \frac{1}{6}A^3A''' \\ &\quad - A(A')^4 - 3A^2(A')^2A'' - \frac{1}{2}A^3(A'')^2 - \frac{2}{3}A^3A'A''' - \frac{1}{24}A^4A'''' + \dots \end{aligned} \quad (\text{B.1})$$

The first line after $x - A$ is the $r = 2$ Lagrange term, the next is $r = 3$, and so on. For example,

$$\frac{1}{24}D^3(A^4) = A(A')^3 + \frac{3}{2}A^2A'A'' + \frac{1}{6}A^3A''',$$

while

$$\begin{aligned} -\frac{1}{120}D^4(A^5) &= -A(A')^4 - 3A^2(A')^2A'' - \frac{1}{2}A^3(A'')^2 \\ &\quad - \frac{2}{3}A^3A'A''' - \frac{1}{24}A^4A'''. \end{aligned}$$

These formulas are excellent for low-order symbolic work, but at high order the compact derivative form should be retained to avoid expression swell.

B.2 A substitution certificate for a truncated inverse

Proposition B.1 (Finite certificate). *Let $F = x + A$ with $A \in \mathcal{P}_{\geq 0}$, and let*

$$G_N = x + \sum_{j=0}^N b_j(\ell) t^j.$$

Then G_N is the truncation through level N of $F^{(-1)}$ if and only if

$$F \circ G_N - x \in t^{N+1} \mathbb{K}[\ell][[t]]. \quad (\text{B.2})$$

The equivalent left certificate is $G_N \circ F - x \in t^{N+1} \mathbb{K}[\ell][[t]]$.

Proof. Necessity follows by truncating the exact identity. Conversely, let G be the unique inverse supplied by the near-identity group theorem. If $G_N - G$ had first nonzero term $q(\ell)t^m$ with $m \leq N$, then Taylor expansion of F at G would give

$$F \circ G_N - F \circ G = (F' \circ G)q(\ell)t^m + O(t^{m+1}).$$

Because $F' \circ G = 1 + O(t)$, the coefficient at level m would be $q \neq 0$, contradicting (B.2). Hence $G_N - G \in t^{N+1} \mathcal{P}_{\geq 0}$. $\square$

This proposition is the simplest robust test for software. It avoids comparing the output with a second implementation term by term: compose once, truncate, and require an exact zero polynomial at every retained level.

B.3 Ramified reversal after dominant balance

Near-identity reversal is only one normal form. Suppose more generally that

$$F(y) = cy^q(1 + U(y)), \quad c \neq 0, \quad U(y) \prec 1, \quad (\text{B.3})$$

where $q \in \mathbb{Q}_{>0}$. Choose a branch of $d = c^{-1/q}$ and write

$$y = dx^{1/q}(1 + V). \quad (\text{B.4})$$

Substitution into (B.3) gives

$$(1 + V)^q(1 + U(dx^{1/q}(1 + V))) = 1. \quad (\text{B.5})$$

The derivative of the left side with respect to V has scalar leading value q , so the formal implicit-function recursion is triangular. The correct ambient ring is generated by $x^{-1/q}$ and the logarithms created by composition. Thus a branch point does not destroy reversibility; it changes the exponent group and makes branch selection part of the leading data.

For $q = 1$, this normalization reduces to the monomial-leading composition calculus. For $q = 2$, it explains the half-integer expansion of the inverse of $y^2 + y$. If q is irrational, a Hahn group containing $q^{-1}\mathbb{Z}$ is required rather than an ordinary Puiseux group.

B.4 Direct implicit recursion

Consider an equation

$$\Psi(x, Y) = 0, \quad Y = Y_0 + \sum_{n \geq 1} y_n(\ell)t^n, \quad (\text{B.6})$$

where all operations in Ψ are admissible and $\partial_Y \Psi(x, Y_0)$ has a nonzero scalar leading coefficient. Suppose $y_1, \dots, y_{N-1}$ have already been determined. Inserting $Y_{<N} + q(\ell)t^N$ changes the residual at its first undecided level by

$$(\partial_Y \Psi(x, Y_0))_{\text{lead}} q(\ell)t^N. \quad (\text{B.7})$$

All nonlinear occurrences of qt^N are of higher level. Therefore the level- N equation is linear in q , and division by the scalar leading coefficient determines it uniquely. This is the coefficient-level content of the filtered implicit-function principle and explains why triangular matching works far beyond literal compositional inverses.

C Lambert-polynomial identities and checks

C.1 Coefficient triangle

Write

$$L_n(X) = \sum_{m=1}^n \lambda_{n,m} X^m, \quad n \geq 1. \quad (\text{C.1})$$

The integral recurrence gives

$$\lambda_{n+1,m} = -\lambda_{n,m} + \frac{n}{m}\lambda_{n,m-1}, \quad (\text{C.2})$$

where $\lambda_{n,0} = 0$. The boundary data are

$$\lambda_{n,1} = (-1)^{n+1}, \quad \lambda_{n,n} = \frac{1}{n}. \quad (\text{C.3})$$

Equivalently,

$$\lambda_{n,m} = \frac{(-1)^{n+1}}{m!} s(n, n-m+1). \quad (\text{C.4})$$

Thus the entire table can be generated using only exact integer arithmetic followed by division by $m!$.

C.2 A differential identity

Differentiating the generating equation

$$\log(1 + T\mathcal{L}) + \mathcal{L} = -X$$

with respect to X gives

$$\partial_X \mathcal{L} = -\frac{1 + T\mathcal{L}}{1 + T + T\mathcal{L}}. \quad (\text{C.5})$$

Together with the T -equation

$$T\mathcal{L}\partial_T \mathcal{L} + (1 + T)\partial_T \mathcal{L} + \mathcal{L} = 0,$$

this gives independent recurrences for derivatives and coefficients. In a computer algebra system, checking both partial differential identities is a stronger test than checking the first few expanded polynomials.

At the level of ω , implicit differentiation of $\omega + \log \omega = x$ yields

$$\omega'(x) = \frac{\omega(x)}{1 + \omega(x)}. \quad (\text{C.6})$$

Substitution of $\omega = x + \sum L_n(\ell)x^{-n}$ into (C.6) reproduces the Lambert recurrence after coefficient collection. Hence the inverse equation, the differential equation, and the Stirling formula are three mutually independent certificates for the same polynomial family.

C.3 Scaling law

The inverse of

$$y + a \log y = x, \quad a \neq 0, \quad (\text{C.7})$$

can be reduced exactly to Wright omega. Put $y = au$; then

$$u + \log u = \frac{x}{a} - \log a,$$

with a consistent logarithm of a . Consequently,

$$y = a \omega\left(\frac{x}{a} - \log a\right). \quad (\text{C.8})$$

Expanding the right side in the original variable x recovers the powers $a^{n+1}L_n(\log x)$ at their highest logarithmic degree, together with the lower-degree corrections caused by the shift $-\log a$ and by re-expanding $\log(x/a - a \log a)$. This identity is a useful exact benchmark for generic reversion code with symbolic parameters.

C.4 Why one should retain the nested variables

For direct evaluation of $W_k(z)$, it is usually preferable to retain

$$\xi_k = \text{Log } z + 2\pi i k, \quad \eta_k = \log \xi_k,$$

and evaluate

$$W_k(z) \approx \xi_k + \sum_{n=0}^N \frac{L_n(\eta_k)}{\xi_k^n}$$

rather than fully expand in powers of $(\text{Log } z)^{-1}$ and $2\pi i k$. The nested form preserves branch information, has smaller intermediate expressions, and permits Horner evaluation of each L_n . A further re-expansion should be performed only when a downstream calculation requires a single prescribed monomial scale.

D Notation and operation tables

D.1 Core notation

Table 2: Notation used throughout the article.

Symbol	Meaning
$t = x^{-1}$	small inverse-power variable as $x \rightarrow \infty$
$\ell = \log x$	first logarithmic scale
$\mathcal{P} = \mathbb{K}[\ell]((t))$	polynomial-logarithmic Laurent-series ring
$\text{ord}_{x^{-1}} F$	least exponent of t occurring in F
$\deg_{\ell} p$	degree of a coefficient polynomial in ℓ
$\Delta_{n,r}$	$\prod_{q=0}^{r-1} (\partial_{\ell} - n - q)$, the shifted derivative operator
$F^{\langle -1 \rangle}$	compositional inverse of F , not multiplicative reciprocal
$L_n(X)$	Lambert polynomial at inverse-power level n
$s(n, k)$	signed Stirling number of the first kind
$\left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right]$	unsigned Stirling cycle number
$\sim$	Poincaré asymptotic expansion in the stated scale
$O(t^N \ell^d)$	analytic remainder bound when explicitly justified

D.2 Closure table

Table 3: Closure of the basic PL class and the usual repair when closure fails.

Operation	Closed in $\mathbb{K}[\ell]((t))$?	Condition or minimal enlargement
Addition, subtraction	yes	no qualification
Multiplication	yes	Cauchy convolution over $\mathbb{K}[\ell]$
Differentiation	yes	$D(t^n p) = t^{n+1}(\partial_{\ell} - n)p$
Antidifferentiation	yes	solve a first-order polynomial equation at each level
Reciprocal	only for units	scalar leading coefficient; otherwise use $\mathbb{K}(\ell)((t))$ or a Laurent-log scale
Integer powers	yes for units and finite non-negative powers	negative powers require a unit

Operation	Closed in $\mathbb{K}[\ell]((t))$?	Condition or minimal enlargement
Rational powers	conditionally	scalar-leading unit after extracting a monomial; use Puiseux exponents if needed
Logarithm	conditionally	extract cx^q ; a leading ℓ^d creates $\log \ell$
Exponential	rarely	allowed for infinitesimal arguments; a nonzero polynomial in ℓ may change the monomial group
Composition	conditionally	inner series must be large with controlled monomial leading term
Near-identity reversal	yes	$F = x + A$, $\text{ord}_{x^{-1}} A \geq 0$
General reversal	conditionally	determine dominant balance; often Puiseux-log or Hahn enlargement

D.3 A minimal implementation checklist

A reliable finite-precision implementation should enforce the following invariants.

1. Store each coefficient in canonical polynomial form and omit exact zero levels.
2. Attach the exponent group and coefficient domain to every series value; never silently coerce $\mathbb{K}[\ell]$ to $\mathbb{K}(\ell)$ or $\mathbb{Z}$ exponents to $\mathbb{Q}$ exponents.
3. Normalize a nonzero series into leading monomial, leading coefficient, and scalar-leading unit before applying reciprocal, logarithm, or a rational power.
4. Propagate a target truncation through every product, derivative, and composition. Do not form terms that provably lie beyond the target.
5. Carry logarithm branches and fractional-power roots as explicit metadata in complex calculations.
6. Verify division by multiplying back, verify composition with an independently truncated direct substitution, and verify reversal using [Theorem B.1](#).
7. For numerical use, estimate terms at the actual argument and truncate near the smallest term unless a convergence theorem applies.

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